

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

MONOETHANOLAMINE (MEA)

Version 2

Revision Date 30.04.2019

Print Date 05.11.2019

GB / EN

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier

Trade name : MONOETHANOLAMINE (MEA)

Index-No. : 603-030-00-8

REACH Registration Number : 01-2119486455-28-0004

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Substance/Mixture : Specific use(s): Refer to attached exposure scenario Annex.

1.3 Details of the supplier of the safety data sheet

Company : Nouryon
Functional Chemicals AB
SE 444 85 Stenungsund
Sweden

Telephone : +4630385000
Telefax : +46303770551
E-mail address : QTS@nouryon.com

1.4 Emergency telephone number

Emergency telephone number : 24 hours emergency response number: +31 57 06 79211
Kemiakuten-SE: 020 99 60 00

SECTION 2: HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

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Acute toxicity, 4, H302
Acute toxicity, 4, H332
Acute toxicity, 4, H312
Skin corrosion, 1B, H314
Serious eye damage, 1, H318
Specific target organ toxicity - single exposure, 3, Respiratory system, H335
Long-term (chronic) aquatic hazard, 3, H412

For the full text of the H-Statements mentioned in this Section, see Section 16.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

Pictogram

:



Signal word

: Danger

Hazard statements

: H302 + H312 + H332 Harmful if swallowed, in contact with skin or if inhaled.
H314 Causes severe skin burns and eye damage.
H335 May cause respiratory irritation.
H412 Harmful to aquatic life with long lasting effects.

Precautionary statements

: **Prevention:**
P261 Avoid breathing mist, vapours or spray.
P273 Avoid release to the environment.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.
Response:
P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water.
P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/doctor.

Hazardous components which must be listed on the label:

Ethanolamine

141-43-5

2.3 Other hazards

No further data available.

PBT and vPvB assessment : This substance/mixture contains no components considered

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to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

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SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Substances

Pure substance/mixture : Substance
CAS-No. : 141-43-5

Hazardous substance

Chemical name	PBT vPvB OEL	CAS-No. EC-No. REACH No.	Classification (REGULATION (EC) No 1272/2008)	Concentration [%]
Ethanolamine		141-43-5 205-483-3 01-2119486455-28	Acute Tox. 4; H302 Acute Tox. 4; H332 Acute Tox. 4; H312 Skin Corr. 1B; H314 Eye Dam. 1; H318 STOT SE 3; H335 Aquatic Chronic 3; H412	>= 90 - <= 100

For the full text of the H-Statements mentioned in this Section, see Section 16.

REACH - Candidate List of Substances of Very High Concern for Authorisation (Article 59).

Status : Not applicable

SECTION 4: FIRST AID MEASURES

4.1 Description of first aid measures

General advice : Immediate medical attention is required.
Move out of dangerous area.
Show this safety data sheet to the doctor in attendance.

If inhaled : If breathed in, move person into fresh air.
Consult a physician after significant exposure.

In case of skin contact : Take off contaminated clothing and shoes immediately.
Rinse immediately with plenty of water.
Immediate medical treatment is necessary as untreated wounds from corrosion of the skin heal slowly and with difficulty.
If skin irritation persists, call a physician.

In case of eye contact : Rinse with plenty of water.
Get medical attention immediately. Continue to rinse during transport.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
Small amounts splashed into eyes can cause irreversible tissue damage and blindness.

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If swallowed : Clean mouth with water and drink afterwards plenty of water.
Never give anything by mouth to an unconscious person.
Take victim immediately to hospital.
Do not induce vomiting! May cause chemical burns in mouth and throat.

4.2 Most important symptoms and effects, both acute and delayed

Symptoms : The symptoms and effects are as expected from the hazards as shown in section 2. No specific product related symptoms are known.

Risks : Harmful if swallowed, in contact with skin or if inhaled.
Causes serious eye damage.
May cause respiratory irritation.
Causes severe burns.

4.3 Indication of any immediate medical attention and special treatment needed

Treatment : Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1 Extinguishing media

Suitable extinguishing media : Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

5.2 Special hazards arising from the substance or mixture

Specific hazards during firefighting / Specific hazards arising from the chemical : Water spray may be ineffective unless used by experienced firefighters.
Do not allow run-off from fire fighting to enter drains or water courses.

Combustion products : Carbon oxides
Nitrogen oxides (NOx)

5.3 Advice for firefighters

Special protective equipment for firefighters : In the event of fire, wear self-contained breathing apparatus.
Further information : Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
For safety reasons in case of fire, cans should be stored separately in closed containments.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Use personal protective equipment.
Wear respiratory protection.
Ensure adequate ventilation.

Emergency measures on : Evacuate personnel to safe areas.

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accidental release

Only qualified personnel equipped with suitable protective equipment may intervene.
Prevent unauthorised persons entering the zone.

6.2 Environmental precautions

Environmental precautions

: Try to prevent the material from entering drains or water courses.
If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and materials for containment and cleaning up

Methods for cleaning up /
Methods for containment

: Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).

6.4 Reference to other sections

For disposal considerations see section 13.
For personal protection see section 8.

SECTION 7: HANDLING AND STORAGE

7.1 Precautions for safe handling

Advice on safe handling

: For personal protection see section 8.
Avoid formation of aerosol.
Do not breathe vapours or spray mist.
Avoid contact with skin, eyes and clothing.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
Dispose of rinse water in accordance with local and national regulations.

Advice on protection against
fire and explosion

: Avoid formation of aerosol.
Keep away from sources of ignition - No smoking.
No sparking tools should be used.
Take measures to prevent the build up of electrostatic charge.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage
areas and containers

: No smoking.
Keep container tightly closed in a dry and well-ventilated place.
Reacts with copper, aluminium, zinc and their alloys.

Other data

: No decomposition if stored and applied as directed.

7.3 Specific end use(s)

Specific use(s)

: Refer to attached exposure scenario Annex.

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SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Components with workplace control parameters

Components	CAS-No.	Value	Control parameters	Update	Basis	Form of exposure
Ethanolamine	141-43-5	TWA	1 ppm 2.5 mg/m ³	2006-02-09	2006/15/EC	
	Further information	:	Indicative skin: Identifies the possibility of significant uptake through the skin			
		STEL	3 ppm 7.6 mg/m ³	2006-02-09	2006/15/EC	
	Further information	:	Indicative skin: Identifies the possibility of significant uptake through the skin			
		TWA	1 ppm 2.5 mg/m ³	2007-08-01	GB EH40	
	Further information	:	Sk: Can be absorbed through skin. The assigned substances are those for which there are concerns that dermal absorption will lead to systemic toxicity.			
		STEL	3 ppm 7.6 mg/m ³	2007-08-01	GB EH40	
	Further information	:	Sk: Can be absorbed through skin. The assigned substances are those for which there are concerns that dermal absorption will lead to systemic toxicity.			

ACGIH: American Conference of Governmental Industrial Hygienists
AGW: Arbeitsplatzgrenzwert
BEI: Biological Exposure Index
MAC: Maximum Allowable Concentration
NIOSH: National Institute for Occupational Safety and Health
OEL: OEL: Occupational exposure limit.
STEL: Short term exposure limit
TRGS: Technische Regel für Gefahrstoffe
TWA: Time Weighted Average

Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006

Substance name	End Use	Exposure routes	Potential health effects	Value
Ethanolamine	Workers	Skin contact	Long-term systemic effects	1 mg/kg bw /day
	Workers	Inhalation	Long-term local effects	3.3 mg/m ³
	Consumers	Skin contact	Long-term systemic effects	0.24 mg/kg bw /day
	Consumers	Ingestion	Long-term systemic effects	3.75 mg/kg bw /day
	Consumers	Inhalation	Long-term local effects	2 mg/m ³

Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006

Substance name	Environmental Compartment	Value
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Ethanolamine	Fresh w ater	0.085 mg/l
	Marine w ater	0.0085 mg/l
	Sediment	0.434 mg/kg dry w eight
	Marine sediment	0.0434 mg/kg dry w eight
	Soil	0.0367 mg/kg dry w eight
	Sew age treatment plant	100 mg/l
	Intermittent w ater	0.028 mg/l

8.2 Exposure controls

Engineering controls

Effective exhaust ventilation system

Ensure that eyewash stations and safety showers are close to the workstation location.

Personal protective equipment

Respiratory protection : In the case of vapour or aerosol formation use a respirator with an approved filter.
Wear full face mask supplied with:
Gas cartridge K (ammonia, green).

Hand protection : butyl-rubber
Break through time: > 30 min
Glove thickness: >= 0.2 mm
Wearing time
< 30 minutes

butyl-rubber
Break through time: > 240 min
Glove thickness: >= 0.6 mm
Wearing time
< 240 minutes

butyl-rubber
Break through time: > 480 min
Glove thickness: >= 0.8 mm
Wearing time
< 480 minutes

The data about break through time/strength of material are standard values! The exact break through time/strength of material has to be obtained from the producer of the protective glove.

Eye protection : Safety glasses with side-shields conforming to EN166

Skin and body protection : Protective suit

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice.
When using do not eat or drink.
When using do not smoke.
Wash hands before breaks and at the end of workday.

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Environmental exposure controls

General advice : Try to prevent the material from entering drains or water courses.
If the product contaminates rivers and lakes or drains inform respective authorities.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Appearance

Form : Clear liquid
Colour : colourless
Odour : amine-like
Odour Threshold : No data available

Safety data

pH : 12.1
at 20 °C
Melting point/freezing point : 4 °C
at 1,013 hPa
Boiling point/boiling range : 167 °C
at 1,013 hPa
Flash point : 92.5 °C
Method: ISO 2719
Evaporation rate : No data available
Flammability (solid, gas) : Not applicable
Flammability (liquids) : Product is combustible at high temperatures.
Lower explosion limit : 3.4 %(V)
at 88.3 °C
Upper explosion limit : 27.0 %(V)
at 133.8 °C
Vapour pressure : 0.5 hPa at 20 °C
Relative vapour density : 2.1
Density : 1,016 kg/m³ at 20 °C
1,000 kg/m³ at 40 °C
Relative density : 1.016 at 20 °C

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Water solubility	: completely miscible
Solubility in other solvents	: Soluble in ethanol and acetone.
Partition coefficient: n-octanol/water	: log Pow: -2.3 at 25 °C
Auto-ignition temperature	: 424 °C at 1,013 hPa
Decomposition temperature	: No data available
Viscosity, dynamic	: 23.18 mPa.s at 20 °C
Viscosity, kinematic	: 23.55 mm ² /s at 20 °C
Explosive properties	: Not explosive
Oxidizing properties	: The substance or mixture is not classified as oxidizing.

9.2 Other information

Molecular weight : 61.08 g/mol

This safety datasheet only contains information relating to safety and does not replace any product information or product specification.

SECTION 10: STABILITY AND REACTIVITY

10.1 Reactivity

Stable under normal conditions.

10.2 Chemical stability

Stable under recommended storage conditions.

10.3 Possibility of hazardous reactions

Heating can release hazardous gases.

10.4 Conditions to avoid

Conditions to avoid : Heat, flames and sparks.

10.5 Incompatible materials

Materials to avoid : Reacts with copper, aluminium, zinc and their alloys.
Strong acids and oxidizing agents
Halogenated compounds

10.6 Hazardous decomposition products

Hazardous decomposition products : Nitrogen oxides (NO_x)

Thermal decomposition : No data available

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SECTION 11: TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Product information:

Acute toxicity	:	Harmful if swallowed, in contact with skin or if inhaled.
Skin corrosion/irritation	:	Causes severe burns.
Serious eye damage/eye irritation	:	Causes serious eye damage.
Respiratory or skin sensitisation	:	Respiratory sensitisation: Not classified based on available information. Skin sensitisation: Not classified based on available information.
Germ cell mutagenicity	:	Not classified based on available information.
Carcinogenicity	:	Not classified based on available information.
Reproductive toxicity	:	Not classified based on available information.
STOT - single exposure	:	May cause respiratory irritation.
STOT - repeated exposure	:	Not classified based on available information.
Aspiration hazard	:	Not classified based on available information.
Further information	:	No further data available.

Toxicology data for the components:

Ethanolamine

Acute toxicity:

Acute oral toxicity	:	LD50: 1,089 mg/kg Species: Rat Method: OECD Test Guideline 401 Information taken from reference works and the literature.
Acute inhalation toxicity	:	LC50 : 20 mg/l Exposure time: 4 h Test atmosphere: vapour Method: Acute toxicity estimate
Acute dermal toxicity	:	LD50: 2,000 mg/kg Method: Acute toxicity estimate
Skin corrosion/irritation	:	Species: Rabbit Result: Causes burns. Method: OECD Test Guideline 404 Information taken from reference works and the literature.
Serious eye damage/eye irritation	:	Species: Rabbit Result: Risk of serious damage to eyes. Method: OECD Test Guideline 405 Information taken from reference works and the literature.

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Respiratory or skin sensitisation	: Maximisation Test Species: Guinea pig Result: Does not cause skin sensitisation. Information taken from reference works and the literature.
Germ cell mutagenicity	
Genotoxicity in vitro	: Ames test Salmonella typhimurium Result: negative Method: OECD Test Guideline 471 Information taken from reference works and the literature.
Genotoxicity in vivo	: Species: Mouse Method: Mutagenicity (micronucleus test) Result: negative Information taken from reference works and the literature.
Carcinogenicity	: study scientifically unjustified
Reproductive toxicity	: Not classified due to data which are conclusive although insufficient for classification.
STOT - single exposure	: May cause respiratory irritation.
STOT - repeated exposure	: Not classified due to data which are conclusive although insufficient for classification.
Aspiration hazard	: Not classified due to data which are conclusive although insufficient for classification.

SECTION 12: ECOLOGICAL INFORMATION

Product information:

Ecotoxicology Assessment

Additional ecological information	: An environmental hazard cannot be excluded in the event of unprofessional handling or disposal. Toxic to aquatic life. Harmful to aquatic life with long lasting effects.
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12.1 Toxicity

Components:

Test result

Ethanolamine

Toxicity to fish	: LC50: 349 mg/l Exposure time: 96 h Species: Cyprinus carpio (Carp) Test Type: semi-static test Method: Tested according to Directive 92/69/EEC. Information taken from reference works and the literature.
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Toxicity to daphnia and other aquatic invertebrates : EC50: 65 mg/l
Exposure time: 48 h
Species: *Daphnia magna* (Water flea)
Test Type: static test
Method: 84/449/EEC C.2
Information taken from reference works and the literature.

Toxicity to algae : EC50: 2.8 mg/l
Exposure time: 72 h
Species: *Pseudokirchneriella subcapitata* (green algae)
Method: OECD Test Guideline 201
Information taken from reference works and the literature.

Toxicity to fish (Chronic toxicity) : NOEC: 1.2 mg/l
Exposure time: 30 d
Species: *Oryzias latipes* (Orange-red killifish)
Method: OECD Test Guideline 210
Information taken from reference works and the literature.

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC: 0.85 mg/l
Exposure time: 21 d
Species: *Daphnia magna* (Water flea)
Method: OECD Test Guideline 202
Information taken from reference works and the literature.

12.2 Persistence and degradability

Product information:

Biodegradability : Result: Readily biodegradable.

Components:

Ethanolamine

Biodegradability : Result: Readily biodegradable.
Method: OECD Test Guideline 301E

12.3 Bioaccumulative potential

Product information:

Bioaccumulation : Not expected considering the low log Pow value.

Components:

Ethanolamine

Bioaccumulation : Not expected considering the low log Pow value.

12.4 Mobility in soil

Product information:

Mobility : Adsorption to the solid soil particles is not expected.
Transport to air is not expected.

Components:

Ethanolamine

Mobility : Mobile in soils
Transport to air is not expected.

12.5 Results of PBT and vPvB assessment

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Product information:

PBT and vPvB assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Components:

Ethanolamine

PBT and vPvB assessment : This substance is not considered to be a PBT (Persistent, Bioaccumulation, Toxic)
This substance is not considered to be vPvB (very Persistent nor very Bioaccumulating)

12.6 Other adverse effects

Product information:

Biochemical Oxygen Demand (BOD) : No data available

Components:

Ethanolamine

Biochemical Oxygen Demand (BOD) : 800 mg/g

SECTION 13: DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Product : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Hazardous waste
Dispose of contents/container in accordance with local regulation.

Contaminated packaging : Empty remaining contents.
Dispose of as unused product.

SECTION 14: TRANSPORT INFORMATION

14.1 UN number

ADR : UN 2491
RID : UN 2491
IMDG-Code : UN 2491
IATA-DGR : UN 2491

14.2 Proper shipping name

ADR : ETHANOLAMINE
RID : ETHANOLAMINE
IMDG-Code : ETHANOLAMINE
IATA-DGR : Ethanolamine

14.3 Transport hazard class

ADR : 8
RID : 8
IMDG-Code : 8
IATA-DGR : 8

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14.4 Packing group

ADR

Packing group : III
Classification Code : C7
Hazard Identification Number : 80
Labels : 8
Tunnel restriction code : (E)

RID

Packing group : III
Classification Code : C7
Hazard Identification Number : 80
Labels : 8

IMDG-Code

Packing group : III
Labels : 8
EmS Code : F-A, S-B

IATA-DGR

Packing instruction (cargo aircraft) : 856
Packing instruction (passenger aircraft) : 852
Packing instruction (LQ) : Y841
Packing group : III
Labels : 8

14.5 Environmental hazards

ADR

Environmentally hazardous : no

RID

Environmentally hazardous : no

IMDG-Code

Marine pollutant : no

IATA-DGR

Environmentally hazardous : no

14.6 Special precautions for user

Not applicable

14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

SECTION 15: REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Seveso III: Directive 2012/18/EU of the European Parliament and of the Council on the control of major-accident hazards involving dangerous substances.

Not applicable

Notification status

DSL : YES. All components of this product are on the Canadian DSL
AICS : YES. On the inventory, or in compliance with the inventory
NZIoC : YES. On the inventory, or in compliance with the inventory
ENCS : YES. On the inventory, or in compliance with the inventory

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ISHL	: YES. On the inventory, or in compliance with the inventory
KECI	: YES. On the inventory, or in compliance with the inventory
PICCS	: YES. On the inventory, or in compliance with the inventory
IECSC	: YES. On the inventory, or in compliance with the inventory
TCSI	: YES. On the inventory, or in compliance with the inventory
TSCA	: YES. All chemical substances in this product are either listed on the TSCA Inventory or in compliance with a TSCA Inventory exemption.

For explanation of abbreviation see section 16.

15.2 Chemical safety assessment

Ethanolamine : A Chemical Safety Assessment has been carried out for this substance.

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3.

H302	: Harmful if swallowed.
H312	: Harmful in contact with skin.
H314	: Causes severe skin burns and eye damage.
H318	: Causes serious eye damage.
H332	: Harmful if inhaled.
H335	: May cause respiratory irritation.
H412	: Harmful to aquatic life with long lasting effects.

Full text of other abbreviations

2006/15/EC	: Europe. Indicative occupational exposure limit values
GB EH40	: UK. EH40 WEL - Workplace Exposure Limits
2006/15/EC / TWA	: Limit Value - eight hours
2006/15/EC / STEL	: Short term exposure limit
GB EH40 / TWA	: Long-term exposure limit (8-hour TWA reference period)
GB EH40 / STEL	: Short-term exposure limit (15-minute reference period)

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - European Agreement concerning the International Carriage of Dangerous Goods by Road; AICS - Australian Inventory of Chemical Substances; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect

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Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

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Annex :

Formulation of preparations

Formulation of preparations

Use as an intermediate

Industrial uses, Additive in concrete and cement

Professional uses, Additive in concrete and cement

Gas treatment

Water treatment chemicals

Industrial uses, Use in metal working fluids/rolling oils

Professional uses, Use in metal working fluids/rolling oils

. Industrial uses, Use in electroplating

. Industrial uses, Use as additive in plastic, e.g. rubber

. Professional uses, Use as additive in plastic, e.g. rubber

. Processing aid for paper, textile, leather

. Industrial use in Detergents and cleaners

. Professional use in Detergents and cleaners

. Use of detergents and cleaners

. Use in Personal care products

. Wood protection formulations

. Wood protection formulations

. Wood protection formulations

. Wood protection formulations

. Industrial application of coatings and inks

. Professional application of coatings and inks

. Consumer application of coatings

. Industrial uses, Use in oilfield applications

. Professional uses, Use in oilfield applications

. Industrial uses, Adhesives, sealants

. Professional uses, Adhesives, sealants

. Use in adhesives and sealants (Consumer)

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- . Laboratory Reagents
- . Laboratory Reagents

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1. Short title of Exposure Scenario: Formulation of preparations

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC2: Formulation of preparations
Process categories	: PROC1: Use in closed process, no likelihood of exposure PROC2: Use in closed, continuous process with occasional controlled exposure PROC3: Use in closed batch process (synthesis or formulation) PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

2.1 Contributing scenario controlling environmental exposure for: ERC2: Formulation of preparations

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 10 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 0.285714 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 350
Emission or Release Factor: Air	: 0 %
Emission or Release Factor: Water	: 0.06 %
Emission or Release Factor: Soil	: 0 %
Remarks	: AISE SPERC 2.1.g.v1

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine
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Air	: sediment, Soil, Sewage treatment plant : Waste gas treatment by thermal oxidation, Exhaust air scrubber
Water	: Aerobic biological treatment
Soil	: Sealing of all relevant soil surfaces in the facility is required, Sewage sludge incineration, No application of sewage sludge to soil

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant	: Municipal sewage treatment plant
Percentage removed from waste water	: 99 %

2.2 Contributing scenario controlling worker exposure for: PROC1: Use in closed process, no likelihood of exposure

Product characteristics

Concentration of the Substance in Mixture/Article	: Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use)	: liquid

Frequency and duration of use

Exposure duration	: < 480 min
Remarks	: Inhalation, Dermal
Frequency of use	: <= 240 days/year

Human factors not influenced by risk management

Dermal exposure	: Palm of one hand (240 cm ²)
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Other operational conditions affecting workers exposure

Outdoor / Indoor	: Indoor
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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

Concentration of the Substance in Mixture/Article	: Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use)	: liquid

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Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

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Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

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2.6 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.7 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 1 - 4 h
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

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Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.8 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 95 %)

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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.9 Contributing scenario controlling worker exposure for: PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

Product characteristics

Concentration of the Substance in Mixture/Article	: Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use)	: liquid

Frequency and duration of use

Exposure duration	: < 480 min
Remarks	: Inhalation, Dermal
Frequency of use	: <= 240 days/year

Human factors not influenced by risk management

Dermal exposure	: Palms of both hands (480 cm ²)
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Other operational conditions affecting workers exposure

Outdoor / Indoor	: Indoor
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Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

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3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC2	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001 mg/L	< 0.0001
			Marine sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Soil		< 0.0001 mg/kg dry weight	0.000537
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC1	ECETOC TRA		Long term inhalation	0.02545 mg/m ³	0.00771
	ECETOC TRA		Long term dermal	0.00171 mg/kg bw/day	0.00156
PROC2	ECETOC TRA		Long term inhalation	2.545 mg/m ³	0.77121
	ECETOC TRA		Long term dermal	0.0686 mg/kg bw/day	0.06234
PROC3	ECETOC TRA		Long term inhalation	0.7635 mg/m ³	0.23136
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC4	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.34286 mg/kg bw/day	0.31169
PROC5	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338
PROC8a	ECETOC TRA		Long term inhalation	0.4581 mg/m ³	0.13882
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338

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PROC8b	ECETOC TRA		Long term inhalation	0.19088 mg/m ³	0.05784
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338
PROC9	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.34286 mg/kg bw/day	0.31169

ERC2: Formulation of preparations

PROC1: Use in closed process, no likelihood of exposure

PROC2: Use in closed, continuous process with occasional controlled exposure

PROC3: Use in closed batch process (synthesis or formulation)

PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Formulation of preparations

Main User Groups	: SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
Environmental Release Categories	: ERC8b: Wide dispersive indoor use of reactive substances in open systems
Process categories	: PROC3: Use in closed batch process (synthesis or formulation) PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

2.1 Contributing scenario controlling environmental exposure for: ERC8a: Wide dispersive indoor use of processing aids in open systems

Activity	: Environmental exposure assessment for this scenario is not relevant.
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2.2 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article	: Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use)	: liquid

Frequency and duration of use

Exposure duration	: < 480 min
Remarks	: Inhalation, Dermal
Frequency of use	: <= 240 days/year

Human factors not influenced by risk management

Dermal exposure	: Palm of one hand (240 cm ²)
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Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

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(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear respiratory protection. (Effectiveness (of a measure): 90 %)
Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 1 - 4 h
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear respiratory protection. (Effectiveness (of a measure): 90 %)
Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

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Frequency and duration of use

Exposure duration : 1 - 4 h
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear respiratory protection. (Effectiveness (of a measure): 90 %)

Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 15 - 60 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.7 Contributing scenario controlling worker exposure for: PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

Product characteristics

Concentration of the Substance in Mixture/Article	: Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use)	: liquid

Frequency and duration of use

Exposure duration	: 15 - 60 min
Remarks	: Inhalation, Dermal
Frequency of use	: <= 240 days/year

Human factors not influenced by risk management

Dermal exposure	: Palms of both hands (480 cm ²)
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Other operational conditions affecting workers exposure

Outdoor / Indoor	: Indoor
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Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

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3. Exposure estimation and reference to its source

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC3	ECETOC TRA		Long term inhalation	1.527 mg/m ³	0.46273
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC4	ECETOC TRA		Long term inhalation	0.509 mg/m ³	0.15424
	ECETOC TRA		Long term dermal	0.34286 mg/kg bw/day	0.31169
PROC5	ECETOC TRA		Long term inhalation	0.3054 mg/m ³	0.09255
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338
PROC8a	ECETOC TRA		Long term inhalation	0.53445 mg/m ³	0.16196
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338
PROC8b	ECETOC TRA		Long term inhalation	0.509 mg/m ³	0.15424
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338
PROC9	ECETOC TRA		Long term inhalation	1.018 mg/m ³	0.30849
	ECETOC TRA		Long term dermal	0.34286 mg/kg bw/day	0.31169

PROC3: Use in closed batch process (synthesis or formulation)

PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

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1. Short title of Exposure Scenario: Use as an intermediate

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC6a: Use of intermediate
Process categories	: PROC1: Use in closed process, no likelihood of exposure PROC2: Use in closed, continuous process with occasional controlled exposure PROC3: Use in closed batch process (synthesis or formulation) PROC8a: Transfer of substance or preparation (charging/discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/ to vessels/ large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

2.1 Contributing scenario controlling environmental exposure for: ERC6a: Use of intermediate

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 5 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 200
Emission or Release Factor: Air	: 0 %
Emission or Release Factor: Water	: 0.005 %
Emission or Release Factor: Soil	: 0.01 %

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Air	: Waste gas treatment by thermal oxidation, Exhaust air scrubber
Water	: Aerobic biological treatment
Soil	: Sealing of all relevant soil surfaces in the facility is required, Sewage sludge incineration, No application of sewage sludge

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to soil

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant
Percentage removed from waste water : 99 %

2.2 Contributing scenario controlling worker exposure for: PROC1: Use in closed process, no likelihood of exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

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Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

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In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 1 - 4 h
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

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Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 95 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.7 Contributing scenario controlling worker exposure for: PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC6a	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001 mg/L	< 0.0001
			Marine sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC1	ECETOC TRA		Long term inhalation	0.02545 mg/m ³	0.00771
	ECETOC TRA		Long term dermal	0.00171 mg/kg bw/day	0.00156
PROC2	ECETOC TRA		Long term inhalation	2.545 mg/m ³	0.77121
	ECETOC TRA		Long term	0.06857	0.06234

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			dermal	mg/kg bw/day	
PROC3	ECETOC TRA		Long term inhalation	0.7635 mg/m ³	0.23136
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8a	ECETOC TRA		Long term inhalation	0.4581 mg/m ³	0.13882
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338
PROC8b	ECETOC TRA		Long term inhalation	0.19088 mg/m ³	0.05784
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338
PROC9	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.34286 mg/kg bw/day	0.31169

ERC6a: Use of intermediate

PROC1: Use in closed process, no likelihood of exposure

PROC2: Use in closed, continuous process with occasional controlled exposure

PROC3: Use in closed batch process (synthesis or formulation)

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Industrial uses, Additive in concrete and cement

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC5: Industrial use resulting in inclusion into or onto a matrix
Process categories	: PROC7: Industrial spraying PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC10: Roller application or brushing PROC13: Treatment of articles by dipping and pouring CS100: Production or preparation of articles by tableting, compression, extrusion or pelletisation PROC15: Use as laboratory reagent

2.1 Contributing scenario controlling environmental exposure for: ERC5: Industrial use resulting in inclusion into or onto a matrix

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 4.545 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 220
Emission or Release Factor: Air	: 1.7 %
Emission or Release Factor: Water	: 0 %
Emission or Release Factor: Soil	: 0 %
Remarks	: EFCC SPERC 5.1a.v1

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Air	: Waste gas treatment by thermal oxidation, Exhaust air scrubber
Water	: Aerobic biological treatment
Soil	: Sealing of all relevant soil surfaces in the facility is required, Sewage sludge incineration, No application of sewage sludge

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to soil

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

Conditions and measures related to external treatment of waste for disposal

Waste treatment : Waste gases incinerated., Waste gases treated in scrubbers.

2.2 Contributing scenario controlling worker exposure for: PROC7: Industrial spraying

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 1 %.

Mixture/Article

Physical Form (at time of use) : liquid

Amount used

Application rate : > 3 L/min

Frequency and duration of use

Exposure duration : < 300 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Outdoor

: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear respiratory protection. (Effectiveness (of a measure): 90 %)

Wear suitable face shield.

2.3 Contributing scenario controlling worker exposure for: PROC7: Industrial spraying

Activity : Option 1

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 1 %.

Mixture/Article

Physical Form (at time of use) : liquid

Amount used

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Application rate : > 3 L/min

Frequency and duration of use

Exposure duration : < 80 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
: , Spraying with high compressed air use

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear respiratory protection. (Effectiveness (of a measure): 95 %)

Wear suitable face shield.

2.4 Contributing scenario controlling worker exposure for: PROC7: Industrial spraying

Activity : Option 2

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 1 %.
Mixture/Article
Physical Form (at time of use) : liquid

Amount used

Application rate : > 3 L/min

Frequency and duration of use

Exposure duration : < 240 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
: , Spraying with high compressed air use, Ensure that direction
of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

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Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear respiratory protection. (Effectiveness (of a measure): 95 %)

Wear suitable face shield.

2.5 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Outdoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

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Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.7 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Outdoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

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2.8 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.9 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Outdoor

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Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.10 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.11 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

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Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Outdoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.12 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

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2.13 Contributing scenario controlling worker exposure for: CS100: Production or preparation or articles by tableting, compression, extrusion or pelletisation

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Outdoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.14 Contributing scenario controlling worker exposure for: CS100: Production or preparation or articles by tableting, compression, extrusion or pelletisation

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

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Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.15 Contributing scenario controlling worker exposure for: PROC15: Use as laboratory reagent

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC5	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001	< 0.0001

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				mg/L	
			Marine sediment	< 0.0001 mg/kg dry weight	< 0.0001
			Soil	< 0.0001 mg/kg dry weight	0.00062
			Sewage treatment plant	0 mg/L	0

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC7	ART		Long term inhalation	2.1 mg/m ³	0.63636
	ECETOC TRA		Long term dermal	0.01339 mg/kg bw/day	0.01218
PROC7	ART		Long term inhalation	2.5 mg/m ³	0.75758
	ECETOC TRA		Long term dermal	0.00357 mg/kg bw/day	0.00325
PROC7	ART		Long term inhalation	2.5 mg/m ³	0.75758
	ECETOC TRA		Long term dermal	0.01071 mg/kg bw/day	0.00974
PROC8a	ECETOC TRA		Long term inhalation	0.89075 mg/m ³	0.26992
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8a	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8b	ECETOC TRA		Long term inhalation	0.44538 mg/m ³	0.13496
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8b	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC10	ECETOC TRA		Long term inhalation	0.89075 mg/m ³	0.26992
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC10	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC13	ECETOC TRA		Long term inhalation	0.89075 mg/m ³	0.26992

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	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC13	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
CS100	ECETOC TRA		Long term inhalation	0.44538 mg/m ³	0.13496
	ECETOC TRA		Long term dermal	0.00857 mg/kg bw/day	0.00779
CS100	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.00857 mg/kg bw/day	0.00779
PROC15	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.00086 mg/kg bw/day	0.00078

CS100: Production or preparation or articles by tableting, compression, extrusion or pelletisation

ERC5: Industrial use resulting in inclusion into or onto a matrix

PROC10: Roller application or brushing

PROC13: Treatment of articles by dipping and pouring

PROC15: Use as laboratory reagent

PROC7: Industrial spraying

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
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1. Short title of Exposure Scenario: Professional uses, Additive in concrete and cement

Main User Groups	: SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
Environmental Release Categories	: ERC8f: Wide dispersive outdoor use resulting in inclusion into or onto a matrix
Process categories	: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC10: Roller application or brushing PROC11: Non-industrial spraying PROC13: Treatment of articles by dipping and pouring

2.1 Contributing scenario controlling environmental exposure for: ERC9a: Wide dispersive indoor use of substances in closed systems

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 10 %
Fraction of Regional tonnage used locally:	: 0.2 %
Maximum daily site tonnage (kg/day):	: 0.000548 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 365
Emission or Release Factor: Air	: 0 %
Emission or Release Factor: Water	: 1 %
Emission or Release Factor: Soil	: 3 %
Remarks	: EFCC SPERC 8F.1a.v1

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Water	: Aerobic biological treatment

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant	: Municipal sewage treatment plant
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2.2 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Outdoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

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Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Outdoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 1 - 4 h
Remarks : Inhalation, Dermal

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Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Outdoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.7 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

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Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.

2.8 Contributing scenario controlling worker exposure for: PROC11: Non-industrial spraying

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 1 %.
Physical Form (at time of use) : liquid

Amount used

Application rate : > 3 L/min

Frequency and duration of use

Exposure duration : < 300 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm²).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Outdoor
: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

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Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear respiratory protection. (Effectiveness (of a measure): 90 %)

Wear suitable face shield.

2.9 Contributing scenario controlling worker exposure for: PROC11: Non-industrial spraying

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 1 %.

Physical Form (at time of use) : liquid

Amount used

Application rate : > 3 L/min

Frequency and duration of use

Exposure duration : < 180 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear respiratory protection. (Effectiveness (of a measure): 95 %)

Wear suitable face shield.

2.10 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

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Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Outdoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.11 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

3. Exposure estimation and reference to its source

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Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8f	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001 mg/L	< 0.0001
			Marine sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC5	ECETOC TRA		Long term inhalation	0.89075 mg/m ³	0.26992
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC5	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8a	ECETOC TRA		Long term inhalation	2.227 mg/m ³	0.67481
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8a	ECETOC TRA		Long term inhalation	1.909 mg/m ³	0.57841
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC10	ECETOC TRA		Long term inhalation	2.227 mg/m ³	0.67481
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC10	ECETOC TRA		Long term inhalation	2.227 mg/m ³	0.67481
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC11	ART		Long term inhalation	2.1 mg/m ³	0.63636

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	ECETOC TRA		Long term dermal	0.03348 mg/kg bw/day	0.03044
PROC11	ART		Long term inhalation	2.4 mg/m ³	0.72727
	ECETOC TRA		Long term dermal	0.02009 mg/kg bw/day	0.01826
PROC13	ECETOC TRA		Long term inhalation	0.89075 mg/m ³	0.26992
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC13	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117

ERC8f: Wide dispersive outdoor use resulting in inclusion into or onto a matrix

PROC10: Roller application or brushing

PROC11: Non-industrial spraying

PROC13: Treatment of articles by dipping and pouring

PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

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1. Short title of Exposure Scenario: Gas treatment

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC7: Industrial use of substances in closed systems
Process categories	: PROC1: Use in closed process, no likelihood of exposure PROC2: Use in closed, continuous process with occasional controlled exposure PROC3: Use in closed batch process (synthesis or formulation) PROC8a: Transfer of substance or preparation (charging/discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/ to vessels/ large containers at dedicated facilities

2.1 Contributing scenario controlling environmental exposure for: ERC7: Industrial use of substances in closed systems

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 2.857 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 350
Emission or Release Factor: Air	: 0.1 %
Emission or Release Factor: Water	: 0.06 %
Emission or Release Factor: Soil	: 0 %

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Air	: Waste gas treatment by thermal oxidation, Exhaust air scrubber
Water	: Aerobic biological treatment
Soil	: Sealing of all relevant soil surfaces in the facility is required, Sewage sludge incineration, No application of sewage sludge to soil

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Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC1: Use in closed process, no likelihood of exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

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Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

Use suitable eye protection.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

In case of exposure:

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2.5 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 1 - 4 h
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

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Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 95 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC7	EUSES		Fresh water		< 0.0001 mg/L	0.00029
			Fresh water sediment		0.00013 mg/kg dry weight	0.00029
			Marine water		< 0.0001 mg/L	0.00030
			Marine sediment		< 0.0001 mg/kg dry weight	0.00029
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Workers

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Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC1	ECETOC TRA		Long term inhalation	0.02545 mg/m ³	0.00771
	ECETOC TRA		Long term dermal	0.00171 mg/kg bw/day	0.00156
PROC2	ECETOC TRA		Long term inhalation	2.545 mg/m ³	0.77121
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC3	ECETOC TRA		Long term inhalation	0.7635 mg/m ³	0.23136
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8a	ECETOC TRA		Long term inhalation	0.4581 mg/m ³	0.13882
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338
PROC8b	ECETOC TRA		Long term inhalation	0.19088 mg/m ³	0.05784
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338

ERC7: Industrial use of substances in closed systems

PROC1: Use in closed process, no likelihood of exposure

PROC2: Use in closed, continuous process with occasional controlled exposure

PROC3: Use in closed batch process (synthesis or formulation)

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

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1. Short title of Exposure Scenario: Water treatment chemicals

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC7: Industrial use of substances in closed systems
Process categories	: PROC1: Use in closed process, no likelihood of exposure PROC2: Use in closed, continuous process with occasional controlled exposure PROC3: Use in closed batch process (synthesis or formulation) PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC8a: Transfer of substance or preparation (charging/discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/ to vessels/ large containers at dedicated facilities PROC13: Treatment of articles by dipping and pouring

2.1 Contributing scenario controlling environmental exposure for: ERC7: Industrial use of substances in closed systems

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 4.545 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 220
Emission or Release Factor: Air	: 0 %
Emission or Release Factor: Water	: 0.06 %
Emission or Release Factor: Soil	: 0 %

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Air	: Waste gas treatment by thermal oxidation, Exhaust air scrubber
Water	: Aerobic biological treatment
Soil	: Sealing of all relevant soil surfaces in the facility is required,

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Sewage sludge incineration, No application of sewage sludge to soil

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC1: Use in closed process, no likelihood of exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

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Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in : Covers the percentage of the substance in the product up to

Mixture/Article 100 % (unless stated differently).

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

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Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 1 - 4 h
Remarks : Inhalation, Dermal

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Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.7 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 95 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

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Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.8 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 100 % (unless stated differently).

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 1 - 4 h

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

3. Exposure estimation and reference to its source

Environment

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Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC7	EUSES		Fresh water		< 0.0001 mg/L	0.00029
			Fresh water sediment		0.00013 mg/kg dry weight	0.00029
			Marine water		< 0.0001 mg/L	0.00030
			Marine sediment		< 0.0001 mg/kg dry weight	0.00029
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		0.00017 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC1	ECETOC TRA		Long term inhalation	0.02545 mg/m ³	0.00771
	ECETOC TRA		Long term dermal	0.00171 mg/kg bw/day	0.00156
PROC2	ECETOC TRA		Long term inhalation	2.545 mg/m ³	0.77121
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC3	ECETOC TRA		Long term inhalation	0.7635 mg/m ³	0.23136
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC4	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.34286 mg/kg bw/day	0.31169
PROC8a	ECETOC TRA		Long term inhalation	0.4581 mg/m ³	0.13882
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338
PROC8b	ECETOC TRA		Long term inhalation	0.19088 mg/m ³	0.05784
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338
PROC13	ECETOC TRA		Long term inhalation	0.4581 mg/m ³	0.13882
	ECETOC TRA		Long term dermal	0.68571 mg/kg bw/day	0.62338

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ERC7: Industrial use of substances in closed systems
PROC1: Use in closed process, no likelihood of exposure
PROC13: Treatment of articles by dipping and pouring
PROC2: Use in closed, continuous process with occasional controlled exposure
PROC3: Use in closed batch process (synthesis or formulation)
PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises
PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities
PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Industrial uses, Use in metal working fluids/rolling oils

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)
Process categories	: PROC2: Use in closed, continuous process with occasional controlled exposure PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC10: Roller application or brushing PROC13: Treatment of articles by dipping and pouring PROC17: Lubrication at high energy conditions in metal working operations PROC18: General greasing/lubrication at high kinetic energy conditions

2.1 Contributing scenario controlling environmental exposure for: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 10 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 100
Emission or Release Factor: Air	: 2 %
Emission or Release Factor: Water	: 0.1 %
Emission or Release Factor: Soil	: 0 %
Remarks	: ESVOC SpERC 4.7a.v1

Technical conditions and measures / Organizational measures

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Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Air	: Waste gas treatment by thermal oxidation, Filtration, Adsorption, Exhaust air scrubber
Water	: Oil-water separation, Aerobic biological treatment
Soil	: Sealing of all relevant soil surfaces in the facility is required, Sewage sludge incineration, No application of sewage sludge to soil

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

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Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

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2.5 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 1 - 4 h
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.7 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.8 Contributing scenario controlling worker exposure for: PROC17: Lubrication at high energy conditions in metal working operations

Activity : Application of liquids in high speed processes (large scale)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.

Physical Form (at time of use) : liquid

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Frequency and duration of use

Exposure duration : < 240 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 300 m3
Temperature : > 20 °C
Ventilation rate per hour : 3

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear suitable coveralls to prevent exposure to the skin.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.9 Contributing scenario controlling worker exposure for: PROC18: General greasing/lubrication at high kinetic energy conditions

Activity : Application of liquids in high speed processes (large scale)

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 10%.
Mixture/Article
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 240 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 300 m3
Temperature : > 20 °C
Ventilation rate per hour : 3

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

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Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear suitable coveralls to prevent exposure to the skin.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC4	EUSES		Fresh water		< 0.0001 mg/L	0.00033
			Fresh water sediment		0.00014 mg/kg dry weight	0.00033
			Marine water		< 0.0001 mg/L	0.00033
			Marine sediment		< 0.0001 mg/kg dry weight	0.00033
			Soil		< 0.0001 mg/kg dry weight	0.00063
			Sewage treatment plant		0.0002 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC2	ECETOC TRA		Long term inhalation	0.2545 mg/m ³	0.07712
	ECETOC TRA		Long term dermal	0.00686 mg/kg bw/day	0.00623
PROC5	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC8a	ECETOC TRA		Long term inhalation	2.545 mg/m ³	0.77121

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	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC8b	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC10	ECETOC TRA		Long term inhalation	1.527 mg/m ³	0.46273
	ECETOC TRA		Long term dermal	0.13714 mg/kg bw/day	0.12468
PROC13	ECETOC TRA		Long term inhalation	2.545 mg/m ³	0.77121
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC17	ART		Long term inhalation	2.1 mg/m ³	0.63636
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC18	ART		Long term inhalation	2.1 mg/m ³	0.63636
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117

ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

PROC10: Roller application or brushing

PROC13: Treatment of articles by dipping and pouring

PROC17: Lubrication at high energy conditions in metal working operations

PROC18: General greasing/lubrication at high kinetic energy conditions

PROC2: Use in closed, continuous process with occasional controlled exposure

PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Professional uses, Use in metal working fluids/rolling oils

Main User Groups	: SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
Environmental Release Categories	: ERC8a: Wide dispersive indoor use of processing aids in open systems
Process categories	: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC10: Roller application or brushing PROC13: Treatment of articles by dipping and pouring PROC17: Lubrication at high energy conditions in metal working operations PROC18: General greasing/lubrication at high kinetic energy conditions PROC20: Use of functional fluids in small devices

2.1 Contributing scenario controlling environmental exposure for: ERC8a: Wide dispersive indoor use of processing aids in open systems

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 10 %
Fraction of Regional tonnage used locally:	: 0.05 %
Maximum daily site tonnage (kg/day):	: 0.000137 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 365
Emission or Release Factor: Air	: 1.5 %
Emission or Release Factor: Water	: 5 %
Emission or Release Factor: Soil	: 5 %
Remarks	: ESVOC SpERC 8.7c.v1

Conditions and measures related to municipal sewage treatment plant

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Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

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Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 10%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

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2.5 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 15 - 60 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.

2.6 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

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Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.7 Contributing scenario controlling worker exposure for: PROC17: Lubrication at high energy conditions in metal working operations

Activity : Application of liquids in high speed processes, Option 1

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 10%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 90 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Room size : 100 m3

Temperature : > 20 °C

Ventilation rate per hour : 1

Technical conditions and measures

Ensure fixed capturing hood is used. (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.8 Contributing scenario controlling worker exposure for: PROC17: Lubrication at high energy conditions in metal working operations

Activity : Application of liquids in high speed processes, Option 2

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 10%.

Mixture/Article

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Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 100 m³
Temperature : > 20 °C
Ventilation rate per hour : 1

Technical conditions and measures

Ensure fixed capturing hood is used. (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear respiratory protection. (Effectiveness (of a measure): 90 %)
Use suitable eye protection.

2.9 Contributing scenario controlling worker exposure for: PROC18: General greasing/lubrication at high kinetic energy conditions

Activity : Application of liquids in high speed processes, Option 1

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 90 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 100 m³

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Temperature : > 20 °C
Ventilation rate per hour : 1

Technical conditions and measures

Ensure fixed capturing hood is used. (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.10 Contributing scenario controlling worker exposure for: PROC18: General greasing/lubrication at high kinetic energy conditions

Activity : Application of liquids in high speed processes, Option 2

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Room size : 100 m³

Temperature : > 20 °C

Ventilation rate per hour : 1

Technical conditions and measures

Ensure fixed capturing hood is used. (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

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Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear respiratory protection. (Effectiveness (of a measure): 90 %)
Use suitable eye protection.

2.11 Contributing scenario controlling worker exposure for: PROC20: Use of functional fluids in small devices

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8a	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001 mg/L	< 0.0001
			Marine sediment		< 0.0001 mg/kg dry	< 0.0001

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					weight	
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC5	ECETOC TRA		Long term inhalation	0.06857 mg/m ³	0.509
	ECETOC TRA		Long term dermal	0.06234 mg/kg bw/day	0.15424
PROC8a	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC8b	ECETOC TRA		Long term inhalation	2.545 mg/m ³	0.77121
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC10	ECETOC TRA		Long term inhalation	0.89075 mg/m ³	0.26992
	ECETOC TRA		Long term dermal	0.13714 mg/kg bw/day	0.12468
PROC13	ECETOC TRA		Long term inhalation	2.545 mg/m ³	0.77121
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC17	ART	Option 1	Long term inhalation	2.7 mg/m ³	0.81818
	ECETOC TRA		Long term dermal	0.02571 mg/kg bw/day	0.02338
PROC17	ART	Option 2	Long term inhalation	1.4 mg/m ³	0.42424
	ECETOC TRA		Long term dermal	0.13714 mg/kg bw/day	0.12468
PROC18	ART	Option 1	Long term inhalation	2.7 mg/m ³	0.81818
	ECETOC TRA		Long term dermal	0.01286 mg/kg bw/day	0.01169
PROC18	ART	Option 2	Long term inhalation	1.4 mg/m ³	0.42424
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC20	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.00857 mg/kg bw/day	0.00779

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ERC8a: Wide dispersive indoor use of processing aids in open systems
PROC10: Roller application or brushing
PROC13: Treatment of articles by dipping and pouring
PROC17: Lubrication at high energy conditions in metal working operations
PROC18: General greasing/lubrication at high kinetic energy conditions
PROC20: Use of functional fluids in small devices
PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)
PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities
PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Industrial uses, Use in electroplating

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)
Process categories	: PROC2: Use in closed, continuous process with occasional controlled exposure PROC3: Use in closed batch process (synthesis or formulation) PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing) PROC13: Treatment of articles by dipping and pouring

2.1 Contributing scenario controlling environmental exposure for: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 10 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 100
Emission or Release Factor: Air	: 2 %
Emission or Release Factor: Water	: 0.1 %
Emission or Release Factor: Soil	: 0 %
Remarks	: spERC: ESVOC SpERC 4.7a.v1

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Air	: Waste gas treatment by thermal oxidation, Filtration, Adsorption, Exhaust air scrubber
Water	: Oil-water separation, Aerobic biological treatment

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Soil : Sealing of all relevant soil surfaces in the facility is required,
Sewage sludge incineration, No application of sewage sludge
to soil

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant
Sludge Treatment : Sewage sludge incineration

2.2 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

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Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 30%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

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Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.9 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2.5%
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.
In case of exposure:
Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.
Gloves APF 10 (Effectiveness (of a measure): 90 %)
Use suitable eye protection.

2.11 Contributing scenario controlling worker exposure for: PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2.5%
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

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Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

Gloves APF 10 (Effectiveness (of a measure): 90 %)

Use suitable eye protection.

2.13 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2.5%

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 10 (Effectiveness (of a measure): 90 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

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3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC4	EUSES		Fresh water		< 0.0001 mg/L	0.00084
			Fresh water sediment		0.00036 mg/kg dry weight	0.00084
			Marine water		< 0.0001 mg/L	0.00084
			Marine sediment		< 0.0001 mg/kg dry weight	0.00084
			Soil		< 0.0001 mg/kg dry weight	0.00063
			Sewage treatment plant		0.00063 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC2	ECETOC TRA		Long term inhalation	0.7635 mg/m ³	0.23136
	ECETOC TRA		Long term dermal	0.02057 mg/kg bw/day	0.01870
PROC3	ECETOC TRA		Long term inhalation	2.291 mg/m ³	0.69409
	ECETOC TRA		Long term dermal	0.01029 mg/kg bw/day	0.00935
PROC5	ECETOC TRA		Long term inhalation	0.3818 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.20571 mg/kg bw/day	0.18701
PROC8b	ECETOC TRA		Long term inhalation	0.31813 mg/m ³	0.09640
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC9	ECETOC TRA		Long term inhalation	0.31813 mg/m ³	0.09640
	ECETOC TRA		Long term dermal	0.01714 mg/kg bw/day	0.01558
PROC13	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117

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ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)
PROC13: Treatment of articles by dipping and pouring
PROC2: Use in closed, continuous process with occasional controlled exposure
PROC3: Use in closed batch process (synthesis or formulation)
PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)
PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities
PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Industrial uses, Use as additive in plastic, e.g. rubber

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC5: Industrial use resulting in inclusion into or onto a matrix
Process categories	: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC7: Industrial spraying PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing) PROC10: Roller application or brushing PROC13: Treatment of articles by dipping and pouring CS100: Production or preparation of articles by tableting, compression, extrusion or pelletisation PROC15: Use as laboratory reagent

2.1 Contributing scenario controlling environmental exposure for: ERC5: Industrial use resulting in inclusion into or onto a matrix

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 3.333 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 300
Emission or Release Factor: Air	: 0.1 %
Emission or Release Factor: Water	: 0.7 %
Emission or Release Factor: Soil	: 0.01 %
Remarks	: ESVOC SpERC 9.12b.v1

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine
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Air	sediment, Soil, Sewage treatment plant : Waste gas treatment by thermal oxidation, Exhaust air scrubber
Water	: Aerobic biological treatment
Soil	: Sealing of all relevant soil surfaces in the facility is required, Sewage sludge incineration, No application of sewage sludge to soil

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.
Gloves APF 10 (Effectiveness (of a measure): 90 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC7: Industrial spraying

Activity	: Option 1, Spraying (automatic/robotic)
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Product characteristics

Concentration of the Substance in Mixture/Article	: Covers percentage substance in the product up to 2%.
Physical Form (at time of use)	: liquid

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Amount used

Application rate : > 3 L/min

Frequency and duration of use

Exposure duration : < 420 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 300 m3
Ventilation rate per hour : 3
: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Technical conditions and measures

Ensure fixed capturing hood is used. (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear suitable coveralls to prevent exposure to the skin.
In case of exposure:
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable face shield.

2.4 Contributing scenario controlling worker exposure for: PROC7: Industrial spraying

Activity : Option 2, Manual spray application (liquid products)

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 2%.
Mixture/Article
Physical Form (at time of use) : liquid

Amount used

Application rate : > 3 L/min

Frequency and duration of use

Exposure duration : < 300 min

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Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 300 m3
Ventilation rate per hour : 3
: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.
Wear respiratory protection. (Effectiveness (of a measure): 95 %)

2.5 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

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Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

Gloves APF 10 (Effectiveness (of a measure): 90 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 2%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

Gloves APF 10 (Effectiveness (of a measure): 90 %)

Use suitable eye protection.

2.7 Contributing scenario controlling worker exposure for: PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 2%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

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Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

Gloves APF 10 (Effectiveness (of a measure): 90 %)

Use suitable eye protection.

2.8 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 2%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

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2.9 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.

2.10 Contributing scenario controlling worker exposure for: CS100: Production or preparation of articles by tableting, compression, extrusion or pelletisation

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

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Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

Gloves APF 10 (Effectiveness (of a measure): 90 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.11 Contributing scenario controlling worker exposure for: PROC15: Use as laboratory reagent

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 2%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

Gloves APF 10 (Effectiveness (of a measure): 90 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC5	EUSES		Fresh water		< 0.0001 mg/L	0.00183
			Fresh water		0.00079	0.00183

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			sediment		mg/kg dry weight	
			Marine water		< 0.0001 mg/L	0.00183
			Marine sediment		< 0.0001 mg/kg dry weight	0.00183
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC5	ECETOC TRA		Long term inhalation	0.2545 mg/m ³	0.07712
	ECETOC TRA		Long term dermal	0.02743 mg/kg bw/day	0.02494
PROC7	ART	Spraying (automatic/robotic)	Long term inhalation	2.2 mg/m ³	0.66667
	ECETOC TRA		Long term dermal	0.0375 mg/kg bw/day	0.03409
PROC7	ART	Manual spray application (liquid products)	Long term inhalation	2.1 mg/m ³	0.63636
	ECETOC TRA		Long term dermal	0.02679 mg/kg bw/day	0.02435
PROC8a	ECETOC TRA		Long term inhalation	0.509 mg/m ³	0.15424
	ECETOC TRA		Long term dermal	0.02743 mg/kg bw/day	0.02494
PROC8b	ECETOC TRA		Long term inhalation	0.2545 mg/m ³	0.07712
	ECETOC TRA		Long term dermal	0.02743 mg/kg bw/day	0.02494
PROC9	ECETOC TRA		Long term inhalation	0.2545 mg/m ³	0.07712
	ECETOC TRA		Long term dermal	0.01371 mg/kg bw/day	0.01247
PROC10	ECETOC TRA		Long term inhalation	0.509 mg/m ³	0.15424
	ECETOC TRA		Long term dermal	0.02743 mg/kg bw/day	0.02494
PROC13	ECETOC TRA		Long term inhalation	0.509 mg/m ³	0.15424
	ECETOC TRA		Long term dermal	0.01371 mg/kg bw/day	0.01247
CS100	ECETOC TRA		Long term inhalation	0.2545 mg/m ³	0.07712
	ECETOC TRA		Long term	0.00686	0.00623

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			dermal	mg/kg bw/day	
PROC15	ECETOC TRA		Long term inhalation	0.2545 mg/m ³	0.07712
	ECETOC TRA		Long term dermal	0.00069 mg/kg bw/day	0.00062

CS100: Production or preparation of articles by tableting, compression, extrusion or pelletisation

ERC5: Industrial use resulting in inclusion into or onto a matrix

PROC10: Roller application or brushing

PROC13: Treatment of articles by dipping and pouring

PROC15: Use as laboratory reagent

PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

PROC7: Industrial spraying

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Professional uses, Use as additive in plastic, e.g. rubber

Main User Groups	: SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
Environmental Release Categories	: ERC8f, ERC8c: Wide dispersive outdoor use resulting in inclusion into or onto a matrix, Wide dispersive indoor use resulting in inclusion into or onto a matrix
Process categories	: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC10: Roller application or brushing PROC11: Non-industrial spraying PROC13: Treatment of articles by dipping and pouring CS100: Production or preparation of articles by tableting, compression, extrusion or pelletisation PROC15: Use as laboratory reagent

2.1 Contributing scenario controlling environmental exposure for: ERC8f, ERC8c: Wide dispersive outdoor use resulting in inclusion into or onto a matrix, Wide dispersive indoor use resulting in inclusion into or onto a matrix

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 10 %
Fraction of Regional tonnage used locally:	: 0.2 %
Maximum daily site tonnage (kg/day):	: 0.000548 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 365
Emission or Release Factor: Air	: 15 %
Emission or Release Factor: Water	: 1 %
Emission or Release Factor: Soil	: 0.5 %

Conditions and measures related to municipal sewage treatment plant

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Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.
Gloves APF 10 (Effectiveness (of a measure): 90 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

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Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

Gloves APF 10 (Effectiveness (of a measure): 90 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 2%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

Gloves APF 10 (Effectiveness (of a measure): 90 %)

Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

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Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.

2.6 Contributing scenario controlling worker exposure for: PROC11: Non-industrial spraying

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2%.
Physical Form (at time of use) : liquid

Amount used

Application rate : 0.3 - 3 L/min

Frequency and duration of use

Exposure duration : < 240 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm²).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 100 m³
Ventilation rate per hour : 1
: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

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Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.
Wear respiratory protection. (Effectiveness (of a measure): 95 %)

2.7 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.
Wear respiratory protection. (Effectiveness (of a measure): 90 %)

2.8 Contributing scenario controlling worker exposure for: CS100: Production or preparation or articles by tableting, compression, extrusion or pelletisation

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2%.
Physical Form (at time of use) : liquid

Frequency and duration of use

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Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

Gloves APF 10 (Effectiveness (of a measure): 90 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.9 Contributing scenario controlling worker exposure for: PROC15: Use as laboratory reagent

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 2%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

Gloves APF 10 (Effectiveness (of a measure): 90 %)

Wear suitable coveralls to prevent exposure to the skin.

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Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8f	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001 mg/L	< 0.0001
			Marine sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC5	ECETOC TRA		Long term inhalation	0.509 mg/m ³	0.15424
	ECETOC TRA		Long term dermal	0.02743 mg/kg bw/day	0.02494
PROC8a	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.02743 mg/kg bw/day	0.02494
PROC8b	ECETOC TRA		Long term inhalation	0.509 mg/m ³	0.15424
	ECETOC TRA		Long term dermal	0.02743 mg/kg bw/day	0.02494
PROC10	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.02743 mg/kg bw/day	0.02494
PROC11	ART		Long term inhalation	2.1 mg/m ³	0.63636
	ECETOC TRA		Long term dermal	0.05357 mg/kg bw/day	0.04870
PROC13	ECETOC TRA		Long term inhalation	0.0509 mg/m ³	0.01542

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	ECETOC TRA		Long term dermal	0.01371 mg/kg bw/day	0.01247
CS100	ECETOC TRA		Long term inhalation	0.509 mg/m ³	0.15424
	ECETOC TRA		Long term dermal	0.00686 mg/kg bw/day	0.00623
PROC15	ECETOC TRA		Long term inhalation	0.2545 mg/m ³	0.07712
	ECETOC TRA		Long term dermal	0.00069 mg/kg bw/day	0.00062

CS100: Production or preparation of articles by tableting, compression, extrusion or pelletisation

ERC8c: Wide dispersive indoor use resulting in inclusion into or onto a matrix

ERC8f: Wide dispersive outdoor use resulting in inclusion into or onto a matrix

PROC10: Roller application or brushing

PROC11: Non-industrial spraying

PROC13: Treatment of articles by dipping and pouring

PROC15: Use as laboratory reagent

PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Processing aid for paper, textile, leather

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)
Process categories	: PROC2: Use in closed, continuous process with occasional controlled exposure PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC7: Industrial spraying PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing) PROC10: Roller application or brushing PROC13: Treatment of articles by dipping and pouring

2.1 Contributing scenario controlling environmental exposure for: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 12.5 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 80
Emission or Release Factor: Air	: 0.001 %
Emission or Release Factor: Water	: 2 %
Emission or Release Factor: Soil	: 0.01 %

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Air	: Waste gas treatment by thermal oxidation, Exhaust air scrubber

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Water	: Aerobic biological treatment
Soil	: Sealing of all relevant soil surfaces in the facility is required, Sewage sludge incineration, No application of sewage sludge to soil

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

Concentration of the Substance in Mixture/Article	: Covers percentage substance in the product up to 10%.
Physical Form (at time of use)	: liquid

Frequency and duration of use

Exposure duration	: < 480 min
Remarks	: Inhalation, Dermal
Frequency of use	: <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

Product characteristics

Concentration of the Substance in Mixture/Article	: Covers percentage substance in the product up to 10%.
Physical Form (at time of use)	: liquid

Frequency and duration of use

Exposure duration	: < 480 min
Remarks	: Inhalation, Dermal
Frequency of use	: <= 240 days/year

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Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC7: Industrial spraying

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Amount used

Application rate : 0.3 - 3 L/min

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm²).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Room size : 300 m³

Ventilation rate per hour : 3

: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Technical conditions and measures

Ensure fixed capturing hood is used. (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

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Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear suitable coveralls to prevent exposure to the skin.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable face shield.

2.5 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 10%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.7 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 10%.

Mixture/Article

Physical Form (at time of use) : liquid

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Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.8 Contributing scenario controlling worker exposure for: PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

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In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.9 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 10%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 1 - 4 h

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.10 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 10%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

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Outdoor / Indoor

: Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC4	EUSES		Fresh water		0.00159 mg/L	0.01868
			Fresh water sediment		0.00810 mg/kg dry weight	0.01866
			Marine water		0.00016 mg/L	0.01868
			Marine sediment		0.00081 mg/kg dry weight	0.01866
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		0.01580 mg/L	0.00016

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC2	ECETOC TRA		Long term inhalation	0.2545 mg/m ³	0.07712
	ECETOC TRA		Long term dermal	0.00686 mg/kg bw/day	0.00623
PROC4	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC7	ART		Long term inhalation	2.1 mg/m ³	0.63636
	ECETOC TRA		Long term dermal	0.10714 mg/kg bw/day	0.09740

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PROC8a	ECETOC TRA		Long term inhalation	2.545 mg/m ³	0.77121
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC8b	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC9	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC10	ECETOC TRA		Long term inhalation	1.527 mg/m ³	0.46273
	ECETOC TRA		Long term dermal	0.13714 mg/kg bw/day	0.12468
PROC13	ECETOC TRA		Long term inhalation	2.545 mg/m ³	0.77121
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234

ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

PROC10: Roller application or brushing

PROC13: Treatment of articles by dipping and pouring

PROC2: Use in closed, continuous process with occasional controlled exposure

PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

PROC7: Industrial spraying

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
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1. Short title of Exposure Scenario: Industrial use in Detergents and cleaners

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)
Process categories	: PROC3: Use in closed batch process (synthesis or formulation) PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC8a: Transfer of substance or preparation (charging/discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/ to vessels/ large containers at dedicated facilities

2.1 Contributing scenario controlling environmental exposure for: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 4.545 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 220
Emission or Release Factor: Air	: 0 %
Emission or Release Factor: Water	: 100 %
Emission or Release Factor: Soil	: 0 %
Remarks	: AISE SPERC 4.1.v2

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Air	: Waste gas treatment by thermal oxidation, Exhaust air scrubber
Water	: Aerobic biological treatment
Soil	: Sealing of all relevant soil surfaces in the facility is required, Sewage sludge incineration, No application of sewage sludge to soil

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Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

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Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 10%.

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Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC4	EUSES		Fresh water		0.00910 mg/L	0.10704
			Fresh water sediment		0.04640 mg/kg dry weight	0.10692
			Marine water		0.00091 mg/L	0.10704
			Marine sediment		0.00464 mg/kg dry weight	0.10692
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		0.09091 mg/L	0.00091

Workers

Contributing	Exposure Assessment	Specific conditions	Value	Level of	RCR
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Scenario	Method			Exposure	
PROC3	ECETOC TRA		Long term inhalation	0.7635 mg/m ³	0.23136
	ECETOC TRA		Long term dermal	0.00343 mg/kg bw/day	0.00312
PROC4	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8a	ECETOC TRA		Long term inhalation	2.545 mg/m ³	0.77121
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC8b	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234

ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

PROC3: Use in closed batch process (synthesis or formulation)

PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

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1. Short title of Exposure Scenario: Professional use in Detergents and cleaners

Main User Groups	: SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
Environmental Release Categories	: ERC8d: Wide dispersive outdoor use of processing aids in open systems
Process categories	: PROC3: Use in closed batch process (synthesis or formulation) PROC8a: Transfer of substance or preparation (charging/discharging) from/ to vessels/ large containers at non-dedicated facilities PROC10: Roller application or brushing PROC11: Non-industrial spraying PROC13: Treatment of articles by dipping and pouring PROC19: Manual activities involving hand contact

2.1 Contributing scenario controlling environmental exposure for: ERC8d: Wide dispersive outdoor use of processing aids in open systems

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 10 %
Fraction of Regional tonnage used locally:	: 0.075 %
Maximum daily site tonnage (kg/day):	: 0.055 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 365
Emission or Release Factor: Air	: 0 %
Emission or Release Factor: Water	: 100 %
Emission or Release Factor: Soil	: 0 %
Remarks	: AISE SPERC 8a.1.a.v2

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Water	: Aerobic biological treatment

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant	: Municipal sewage treatment plant
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2.2 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 15 - 60 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

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Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 15 - 60 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.5 Contributing scenario controlling worker exposure for: PROC11: Non-industrial spraying

Activity : Option 1, Small scale

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.

Physical Form (at time of use) : liquid

Amount used

Application rate : 0.3 - 3 L/min

Frequency and duration of use

Exposure time : < 60 min

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Frequency of use : minutes/day
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 100 m3
Ventilation rate per hour : 1
: , Spraying with no or low compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.6 Contributing scenario controlling worker exposure for: PROC11: Non-industrial spraying

Activity : Option 2, Small scale

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 10%.

Mixture/Article

Physical Form (at time of use) : liquid

Amount used

Application rate : 0.3 - 3 L/min

Frequency and duration of use

Exposure duration : < 360 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 100 m3
Ventilation rate per hour : 1
: , Spraying with no or low compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

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Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear respiratory protection. (Effectiveness (of a measure): 90 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.

2.7 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.

2.9 Contributing scenario controlling worker exposure for: PROC19: Manual activities involving hand contact

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

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Frequency and duration of use

Exposure duration : < 240 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands and main part of arms (1980 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear respiratory protection. (Effectiveness (of a measure): 90 %)

Wear suitable face shield.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8d	EUSES		Fresh water		0.00036 mg/L	0.00418
			Fresh water sediment		0.00181 mg/kg dry weight	0.00418
			Marine water		< 0.0001 mg/L	0.00418
			Marine sediment		0.00018 mg/kg dry weight	0.00418
			Soil		0.00013 mg/kg dry weight	0.00359
			Sewage treatment plant		0.00348 mg/L	< 0.0001

Workers

Contributing	Exposure Assessment	Specific conditions	Value	Level of	RCR
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Scenario	Method			Exposure	
PROC3	ECETOC TRA		Long term inhalation	0.7635 mg/m ³	0.23136
	ECETOC TRA		Long term dermal	0.00343 mg/kg bw/day	0.00312
PROC8a	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC10	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.13714 mg/kg bw/day	0.12468
PROC11	ART		Long term inhalation	1.8 mg/m ³	0.54546
	ECETOC TRA		Long term dermal	0.06696 mg/kg bw/day	0.06088
PROC11	ART		Long term inhalation	1.1 mg/m ³	0.33333
	ECETOC TRA		Long term dermal	0.40179 mg/kg bw/day	0.36526
PROC13	ECETOC TRA		Long term inhalation	2.545 mg/m ³	0.77121
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC19	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.70714 mg/kg bw/day	0.64286

ERC8d: Wide dispersive outdoor use of processing aids in open systems

PROC10: Roller application or brushing

PROC11: Non-industrial spraying

PROC13: Treatment of articles by dipping and pouring

PROC19: Manual activities involving hand contact

PROC3: Use in closed batch process (synthesis or formulation)

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Use of detergents and cleaners

Main User Groups : SU 21: Consumer uses: Private households (= general public = consumers)
Environmental Release Categories : ERC8d: Wide dispersive outdoor use of processing aids in open systems
Chemical product category : PC35: Washing and cleaning products

2.1 Contributing scenario controlling environmental exposure for: ERC8d: Wide dispersive outdoor use of processing aids in open systems

Amount used

Regional use tonnage : 1 ton(s)/year
(tonnes/year):
Fraction of EU tonnage used in : 10 %
region:
Fraction of Regional tonnage used : 0.075 %
locally:
Maximum daily site tonnage : 0.027 kg/day
(kg/day):

Environment factors not influenced by risk management

Dilution Factor (River) : 10
Dilution Factor (Coastal Areas) : 100

Other given operational conditions affecting environmental exposure

Number of emission days per year : 365
Emission or Release Factor: Air : 0 %
Emission or Release Factor: Water : 100 %
Emission or Release Factor: Soil : 0 %
Remarks : AISE SPERC 8a.1.a.v2

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling consumer exposure for: PC35: Washing and cleaning products

Activity : Detergent, Loading/unloading operations

Product characteristics

Concentration of the Substance in : Covers the percentage of the substance in the product up to
Mixture/Article 5%.
Physical Form (at time of use) : liquid

Amount used

Amount used : 500 g
Remarks : Inhalation
Amount used : 0.010 g

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Remarks : Dermal
Amount used : 0.010 g
Remarks : oral

Frequency and duration of use

Application duration : 0.3 min
Exposure duration : 0.75 min
Remarks : Inhalation, Dermal, Oral exposure

Human factors not influenced by risk management

Dermal exposure : 215 cm²

Other given operational conditions affecting consumers exposure

Outdoor / Indoor : Indoor
Room size : 1 m³
Temperature : 20 °C
Ventilation rate per hour : 2
Remarks : L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Do not touch eyes when using this product.

2.3 Contributing scenario controlling consumer exposure for: PC35: Washing and cleaning products

Activity : Detergent, Application

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 0.05%
Physical Form (at time of use) : liquid

Amount used

Amount used : 19 g
Remarks : Inhalation
Amount used : 19 g
Remarks : Dermal
Amount used : 0.5 g
Remarks : oral

Frequency and duration of use

Application duration : 10 min
Exposure duration : 10 min
Remarks : Inhalation, Dermal, Oral exposure

Other given operational conditions affecting consumers exposure

Outdoor / Indoor : Indoor
Room size : 10 m³
Temperature : 20 °C
Ventilation rate per hour : 2
Remarks : L/h

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Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Do not touch eyes when using this product.

2.4 Contributing scenario controlling consumer exposure for: PC35: Washing and cleaning products

Activity	: All purpose cleaners, Mixing and loading
Product characteristics	
Concentration of the Substance in Mixture/Article	: Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use)	: liquid
Amount used	
Amount used	: 500 g
Frequency and duration of use	
Application duration	: 0.3 min
Exposure duration	: 0.75 min
Remarks	: Inhalation, Dermal, Oral exposure
Human factors not influenced by risk management	
Dermal exposure	: 215 cm ²
Other given operational conditions affecting consumers exposure	
Outdoor / Indoor	: Indoor
Room size	: 1 m ³
Temperature	: 20 °C
Ventilation rate per hour	: 0.5
Remarks	: L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Do not touch eyes when using this product.

2.5 Contributing scenario controlling consumer exposure for: PC35: Washing and cleaning products

Activity	: All purpose cleaners, Application
Product characteristics	
Concentration of the Substance in Mixture/Article	: Covers percentage substance in the product up to 0.06%
Physical Form (at time of use)	: liquid
Amount used	
Amount used	: 400 g
Remarks	: Inhalation
Amount used	: 19 g
Remarks	: Dermal
Amount used	: 0.010 g
Remarks	: oral

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Frequency and duration of use

Application duration : 20 min
Exposure duration : 240 min
Remarks : Inhalation, Dermal, Oral exposure

Human factors not influenced by risk management

Dermal exposure : 1900 cm²

Other given operational conditions affecting consumers exposure

Outdoor / Indoor : Indoor
Room size : 58 m³
Temperature : 20 °C
Ventilation rate per hour : 0.5
Remarks : L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Do not touch eyes when using this product.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8d	EUSES		Fresh water		0.00018 mg/L	0.0021
			Fresh water sediment		0.00091 mg/kg dry weight	0.0021
			Marine water		< 0.0001 mg/L	0.0021
			Marine sediment		< 0.0001 mg/kg dry weight	0.0021
			Soil		< 0.0001 mg/kg dry weight	0.00204
			Sewage treatment plant		0.00171 mg/L	< 0.0001

Consumers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PC35	"Consexpo"	Detergent, Loading/unloading operations	Long term inhalation	< 0.0001 mg/m ³	< 0.0001
	"Consexpo"		Long term dermal	0.00833 mg/kg bw/day	0.03472
	"Consexpo"		Long-term	0.00833	0.00222

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			oral	mg/kg bw/day	
PC35	"Consexpo"	Detergent, Application	Long term inhalation	< 0.0001 mg/m ³	< 0.0001
	"Consexpo"		Long term dermal	0.15833 mg/kg bw/day	0.65972
	"Consexpo"		Long-term oral	0.00417 mg/kg bw/day	0.00111
PC35	"Consexpo"	All purpose cleaners, Mixing and loading	Long term inhalation	< 0.0001 mg/m ³	< 0.0001
	"Consexpo"		Long term dermal	0.00833 mg/kg bw/day	0.03472
	"Consexpo"		Long-term oral	0.00833 mg/kg bw/day	0.00222
PC35	"Consexpo"	All purpose cleaners, Mixing and loading	Long term inhalation	0.03043 mg/m ³	0.01522
	"Consexpo"		Long term dermal	0.1995 mg/kg bw/day	0.83125
	"Consexpo"		Long-term oral	< 0.0001 mg/kg bw/day	< 0.0001

ERC8d: Wide dispersive outdoor use of processing aids in open systems

PC35: Washing and cleaning products

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Use in Personal care products

Main User Groups : SU 21: Consumer uses: Private households (= general public = consumers)
Environmental Release Categories : ERC8a: Wide dispersive indoor use of processing aids in open systems
Chemical product category : PC39: Cosmetics, personal care products

2.1 Contributing scenario controlling environmental exposure for: ERC8a: Wide dispersive indoor use of processing aids in open systems

Amount used

Regional use tonnage : 1 ton(s)/year
(tonnes/year):
Fraction of EU tonnage used in : 10 %
region:
Fraction of Regional tonnage used : 0.075 %
locally:
Maximum daily site tonnage : 0.098 kg/day
(kg/day):

Environment factors not influenced by risk management

Dilution Factor (River) : 10
Dilution Factor (Coastal Areas) : 100

Other given operational conditions affecting environmental exposure

Number of emission days per year : 365
Emission or Release Factor: Air : 0 %
Emission or Release Factor: Water : 100 %
Emission or Release Factor: Soil : 0 %
Remarks : COLIPA SPERC 8a.1.a.v1

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling consumer exposure for: PC39: Cosmetics, personal care products

Activity : Use in Personal care products

3. Exposure estimation and reference to its source

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Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8a	EUSES		Fresh water		0.00063 mg/L	0.00738
			Fresh water sediment		0.00320 mg/kg dry weight	0.00737
			Marine water		< 0.0001 mg/L	0.00738
			Marine sediment		0.00032 mg/kg dry weight	0.00737
			Soil		0.00022 mg/kg dry weight	0.00598
			Sewage treatment plant		0.006195 mg/L	< 0.0001

ERC8a: Wide dispersive indoor use of processing aids in open systems

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Wood protection formulations

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC5: Industrial use resulting in inclusion into or onto a matrix
Process categories	: PROC1: Use in closed process, no likelihood of exposure PROC3: Use in closed batch process (synthesis or formulation) PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing) PROC13: Treatment of articles by dipping and pouring PROC15: Use as laboratory reagent

2.1 Contributing scenario controlling environmental exposure for: ERC5: Industrial use resulting in inclusion into or onto a matrix

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 10 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 100
Emission or Release Factor: Air	: 0 %
Emission or Release Factor: Water	: 0.06 %
Emission or Release Factor: Soil	: 0.01 %

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Air	: Waste gas treatment by thermal oxidation, Exhaust air scrubber

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Water	: Aerobic biological treatment
Soil	: Sewage sludge incineration, No application of sewage sludge to soil, Sealing of all relevant soil surfaces in the facility is required

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant	: Municipal sewage treatment plant
Percentage removed from waste water	: 99 %

2.2 Contributing scenario controlling worker exposure for: PROC1: Use in closed process, no likelihood of exposure

Product characteristics

Concentration of the Substance in Mixture/Article	: Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use)	: liquid

Frequency and duration of use

Exposure duration	: < 480 min
Remarks	: Inhalation, Dermal
Frequency of use	: <= 240 days/year

Human factors not influenced by risk management

Dermal exposure	: Palm of one hand (240 cm ²)
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Other operational conditions affecting workers exposure

Outdoor / Indoor	: Indoor
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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article	: Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use)	: liquid

Frequency and duration of use

Exposure duration	: < 480 min
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Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

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Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

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Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.7 Contributing scenario controlling worker exposure for: PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

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Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.8 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.9 Contributing scenario controlling worker exposure for: PROC15: Use as laboratory reagent

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

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Outdoor / Indoor

: Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC5	EUSES		Fresh water		< 0.0001 mg/L	0.00023
			Fresh water sediment		0.00010 mg/kg dry weight	0.00023
			Marine water		< 0.0001 mg/L	0.00023
			Marine sediment		< 0.0001 mg/kg dry weight	0.00023
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		0.00012 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC1	ECETOC TRA		Long term inhalation	0.00127 mg/m ³	0.00039
	ECETOC TRA		Long term dermal	< 0.0001 mg/kg bw/day	< 0.0001
PROC3	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.00171 mg/kg bw/day	0.00156

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PROC5	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8a	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8b	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC9	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.01714 mg/kg bw/day	0.01558
PROC13	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC15	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.00086 mg/kg bw/day	0.00078

ERC5: Industrial use resulting in inclusion into or onto a matrix

PROC1: Use in closed process, no likelihood of exposure

PROC13: Treatment of articles by dipping and pouring

PROC15: Use as laboratory reagent

PROC3: Use in closed batch process (synthesis or formulation)

PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Wood protection formulations

Main User Groups	: SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
Environmental Release Categories	: ERC8f: Wide dispersive outdoor use resulting in inclusion into or onto a matrix
Process categories	: PROC1: Use in closed process, no likelihood of exposure PROC3: Use in closed batch process (synthesis or formulation) PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC10: Roller application or brushing PROC11: Non-industrial spraying PROC13: Treatment of articles by dipping and pouring

2.1 Contributing scenario controlling environmental exposure for: ERC8f, ERC8c: Wide dispersive outdoor use resulting in inclusion into or onto a matrix, Wide dispersive indoor use resulting in inclusion into or onto a matrix

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 10 %
Fraction of Regional tonnage used locally:	: 0.2 %
Maximum daily site tonnage (kg/day):	: 0.000548 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 365
Emission or Release Factor: Air	: 15 %
Emission or Release Factor: Water	: 1 %
Emission or Release Factor: Soil	: 0.5 %

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant	: Municipal sewage treatment plant
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2.2 Contributing scenario controlling worker exposure for: PROC1: Use in closed process, no likelihood of exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

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Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

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Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

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Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.7 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 1 - 4 h
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.

2.8 Contributing scenario controlling worker exposure for: PROC11: Non-industrial spraying

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Amount used

Application rate : 0.3 - 3 L/min

Frequency and duration of use

Exposure time : < 90 min
Frequency of use : minutes/day
Remarks : Inhalation, Dermal

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Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Room size : 100 m3

Ventilation rate per hour : 1

: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear respiratory protection. (Effectiveness (of a measure): 95 %)

Wear suitable face shield.

2.9 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

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(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.10 Contributing scenario controlling worker exposure for: PROC11: Non-industrial spraying

Activity : Manual process

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Amount used

Application rate : 0.3 - 3 L/min

Frequency and duration of use

Exposure time : < 90 min

Frequency of use : minutes/day

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Room size : 100 m3

Temperature : 20 °C

Ventilation rate per hour : 1

: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear respiratory protection. (Effectiveness (of a measure): 95 %)

Wear suitable face shield.

2.11 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in : Covers the percentage of the substance in the product up to

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Mixture/Article : 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8f	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001 mg/L	< 0.0001
			Marine sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Workers

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Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC1	ECETOC TRA		Long term inhalation	0.00127 mg/m ³	0.00039
	ECETOC TRA		Long term dermal	< 0.0001 mg/kg bw/day	< 0.0001
PROC3	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.00171 mg/kg bw/day	0.00156
PROC4	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.01714 mg/kg bw/day	0.01558
PROC5	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8a	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC10	ECETOC TRA		Long term inhalation	1.909 mg/m ³	0.57841
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC11	ART		Long term inhalation	2 mg/m ³	0.60606
	ECETOC TRA		Long term dermal	0.05022 mg/kg bw/day	0.04566
PROC13	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC11	ART		Long term inhalation	2 mg/m ³	0.60606
	ECETOC TRA		Long term dermal	0.05022 mg/kg bw/day	0.04566
PROC13	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117

ERC8c: Wide dispersive indoor use resulting in inclusion into or onto a matrix

ERC8f: Wide dispersive outdoor use resulting in inclusion into or onto a matrix

PROC1: Use in closed process, no likelihood of exposure

PROC10: Roller application or brushing

PROC11: Non-industrial spraying

PROC13: Treatment of articles by dipping and pouring

PROC3: Use in closed batch process (synthesis or formulation)

PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

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PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Wood protection formulations

Main User Groups : SU 21: Consumer uses: Private households (= general public = consumers)
Environmental Release Categories : ERC8f, ERC8c: Wide dispersive outdoor use resulting in inclusion into or onto a matrix, Wide dispersive indoor use resulting in inclusion into or onto a matrix
Chemical product category : PC8: Biocidal products

2.1 Contributing scenario controlling environmental exposure for: ERC8f, ERC8c: Wide dispersive outdoor use resulting in inclusion into or onto a matrix, Wide dispersive indoor use resulting in inclusion into or onto a matrix

Amount used

Regional use tonnage : 1 ton(s)/year
(tonnes/year):
Fraction of EU tonnage used in : 10 %
region:
Fraction of Regional tonnage used : 0.2 %
locally:
Maximum daily site tonnage : 0.000548 kg/day
(kg/day):

Environment factors not influenced by risk management

Dilution Factor (River) : 10
Dilution Factor (Coastal Areas) : 100

Other given operational conditions affecting environmental exposure

Number of emission days per year : 365
Emission or Release Factor: Air : 15 %
Emission or Release Factor: Water : 1 %
Emission or Release Factor: Soil : 0.5 %
Remarks : ESVOC SpERC 8.16.v1

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling consumer exposure for: PC8: Biocidal products

Amount used

Amount used : 1000 g
Remarks : Inhalation

Frequency and duration of use

Frequency of use : 1 event/day
Remarks : Inhalation, Dermal
Exposure duration : 132 min

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Remarks : Inhalation
Application duration : 120 min
Remarks : Inhalation
Release duration : 120 min
Remarks : Dermal

Human factors not influenced by risk management

Dermal exposure : 860 cm2

Other given operational conditions affecting consumers exposure

Outdoor / Indoor : Indoor
Room size : 900 m3
Temperature : 20 °C
Ventilation rate per hour : 2.5
Remarks : L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Do not touch eyes when using this product.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8f	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001 mg/L	< 0.0001
			Marine sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Consumers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PC8	"Consexpo"		Long term inhalation	0.04714 mg/m3	0.02357
	"Consexpo"		Long term	0.180 mg/kg	0.750

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			dermal	bw/day	
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: Wood protection formulations

ERC8c: Wide dispersive indoor use resulting in inclusion into or onto a matrix

ERC8f: Wide dispersive outdoor use resulting in inclusion into or onto a matrix

PC8: Biocidal products

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Wood protection formulations

Environmental Release Categories : ERC10a: Wide dispersive outdoor use of long-life articles and materials with low release

Article categories : AC11: Wood articles

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC10a	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001 mg/L	< 0.0001
			Marine sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Consumers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
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ERC10a: Wide dispersive outdoor use of long-life articles and materials with low release

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

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1. Short title of Exposure Scenario: Industrial application of coatings and inks

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)
Process categories	: PROC1: Use in closed process, no likelihood of exposure PROC2: Use in closed, continuous process with occasional controlled exposure PROC3: Use in closed batch process (synthesis or formulation) PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC7: Industrial spraying PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC10: Roller application or brushing PROC13: Treatment of articles by dipping and pouring PROC15: Use as laboratory reagent

2.1 Contributing scenario controlling environmental exposure for: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 5 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 225
Emission or Release Factor: Air	: 0.8 %
Emission or Release Factor: Water	: 2 %
Emission or Release Factor: Soil	: 0 %
Remarks	: CEPE SpERC 4

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Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Air	: Waste gas treatment by thermal oxidation, Exhaust air scrubber
Water	: Aerobic biological treatment
Soil	: No application of sewage sludge to soil, Sewage sludge incineration, Sealing of all relevant soil surfaces in the facility is required

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant	: Municipal sewage treatment plant
Percentage removed from waste water	: 99 %

2.2 Contributing scenario controlling worker exposure for: PROC1: Use in closed process, no likelihood of exposure

Product characteristics

Concentration of the Substance in Mixture/Article	: Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use)	: liquid

Frequency and duration of use

Exposure duration	: < 480 min
Remarks	: Inhalation, Dermal
Frequency of use	: <= 240 days/year

Human factors not influenced by risk management

Dermal exposure	: Palm of one hand (240 cm ²)
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Other operational conditions affecting workers exposure

Outdoor / Indoor	: Indoor
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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

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Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

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Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

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Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.7 Contributing scenario controlling worker exposure for: PROC7: Industrial spraying

Activity : Option 1, Spraying (automatic/robotic)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Amount used

Application rate : 0.3 - 3 L/min

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm²).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Room size : 300 m³

Temperature : 20 °C

Ventilation rate per hour : 3

: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Technical conditions and measures

Ensure fixed capturing hood is used. (Effectiveness (of a measure): 90 %)

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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear suitable coveralls to prevent exposure to the skin.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable face shield.

2.8 Contributing scenario controlling worker exposure for: PROC7: Industrial spraying

Activity : Option 2, Manual spray application (liquid products)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Amount used

Application rate : 0.3 - 3 L/min

Frequency and duration of use

Exposure duration : < 360 min

Frequency of use : minutes/day

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Room size : 300 m3

Temperature : 20 °C

Ventilation rate per hour : 3

: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

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Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear respiratory protection. (Effectiveness (of a measure): 95 %)
Wear suitable face shield.

2.9 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.10 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

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Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.11 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

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2.12 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.

2.13 Contributing scenario controlling worker exposure for: PROC15: Use as laboratory reagent

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

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Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC4	EUSES		Fresh water		0.00021 mg/L	0.00244
			Fresh water sediment		0.00106 mg/kg dry weight	0.00244
			Marine water		< 0.0001 mg/L	0.00244
			Marine sediment		0.00011 mg/kg dry weight	0.00244
			Soil		< 0.0001 mg/kg dry weight	0.00058
			Sewage treatment plant		0.002 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC1	ECETOC TRA		Long term inhalation	0.00127 mg/m ³	0.00039
	ECETOC TRA		Long term dermal	< 0.0001 mg/kg bw/day	< 0.0001
PROC2	ECETOC TRA		Long term inhalation	0.12725 mg/m ³	0.03856
	ECETOC TRA		Long term dermal	0.00343 mg/kg bw/day	0.00312
PROC3	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.00171 mg/kg bw/day	0.00156
PROC4	ECETOC TRA		Long term	0.63625	0.19280

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			inhalation	mg/m ³	
	ECETOC TRA		Long term dermal	0.01714 mg/kg bw/day	0.01558
PROC5	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC7	ART	Spraying (automatic/robotic)	Long term inhalation	2.1 mg/m ³	0.63636
	ECETOC TRA		Long term dermal	0.10714 mg/kg bw/day	0.09740
PROC7	ART	Manual spray application (liquid products)	Long term inhalation	2.1 mg/m ³	0.63636
	ECETOC TRA		Long term dermal	0.08036 mg/kg bw/day	0.07305
PROC8a	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8b	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC10	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC13	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC15	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.00086 mg/kg bw/day	0.00078

ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

PROC1: Use in closed process, no likelihood of exposure

PROC10: Roller application or brushing

PROC13: Treatment of articles by dipping and pouring

PROC15: Use as laboratory reagent

PROC2: Use in closed, continuous process with occasional controlled exposure

PROC3: Use in closed batch process (synthesis or formulation)

PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

PROC7: Industrial spraying

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

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For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Professional application of coatings and inks

Main User Groups	: SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
Environmental Release Categories	: ERC8d: Wide dispersive outdoor use of processing aids in open systems
Process categories	: PROC1: Use in closed process, no likelihood of exposure PROC2: Use in closed, continuous process with occasional controlled exposure PROC3: Use in closed batch process (synthesis or formulation) PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing) PROC10: Roller application or brushing PROC11: Non-industrial spraying PROC13: Treatment of articles by dipping and pouring PROC15: Use as laboratory reagent PROC19: Manual activities involving hand contact

2.1 Contributing scenario controlling environmental exposure for: ERC8d: Wide dispersive outdoor use of processing aids in open systems

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 10 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 0.444444 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 225
Emission or Release Factor: Air	: 98 %

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Emission or Release Factor: Water : 2 %
Emission or Release Factor: Soil : 0 %
Remarks : CEPE SPERC 8d.3a.v.1

Technical conditions and measures / Organizational measures

Compartment : Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC1: Use in closed process, no likelihood of exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

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Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

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2.5 Contributing scenario controlling worker exposure for: PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

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Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.7 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.8 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

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Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.9 Contributing scenario controlling worker exposure for: PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

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Use suitable eye protection.

2.10 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.

2.11 Contributing scenario controlling worker exposure for: PROC11: Non-industrial spraying

Activity : Manual process

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Amount used

Application rate : 0.3 - 3 L/min

Frequency and duration of use

Exposure time : < 90 min
Frequency of use : minutes/day
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

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Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 100 m3
Temperature : 20 °C
Ventilation rate per hour : 1
: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear respiratory protection. (Effectiveness (of a measure): 95 %)
Wear suitable face shield.

2.12 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.

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Wear suitable face shield.

2.13 Contributing scenario controlling worker exposure for: PROC15: Use as laboratory reagent

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.14 Contributing scenario controlling worker exposure for: PROC19: Manual activities involving hand contact

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 15 - 60 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands and main part of arms (1980 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8d	EUSES		Fresh water		< 0.0001 mg/L	0.00075
			Fresh water sediment		0.00033 mg/kg dry weight	0.00075
			Marine water		< 0.0001 mg/L	0.00075
			Marine sediment		< 0.0001 mg/kg dry weight	0.00075
			Soil		< 0.0001 mg/kg dry weight	0.00103
			Sewage treatment plant		0.00056 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC1	ECETOC TRA		Long term inhalation	0.00127 mg/m ³	0.00039
	ECETOC TRA		Long term dermal	0.00009 mg/kg bw/day	0.00008
PROC2	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.00343 mg/kg bw/day	0.00312
PROC3	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term	0.00171	0.00156

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			dermal	mg/kg bw/day	
PROC4	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.01714 mg/kg bw/day	0.01558
PROC5	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8a	ECETOC TRA		Long term inhalation	2.227 mg/m ³	0.67481
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8b	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC9	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.01714 mg/kg bw/day	0.01558
PROC10	ECETOC TRA		Long term inhalation	2.227 mg/m ³	0.67481
	ECETOC TRA		Long term dermal	0.06857 mg/kg bw/day	0.06234
PROC11	ART		Long term inhalation	2 mg/m ³	0.60606
	ECETOC TRA		Long term dermal	0.05022 mg/kg bw/day	0.04566
PROC13	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC15	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.00086 mg/kg bw/day	0.00078
PROC19	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.35357 mg/kg bw/day	0.32143

ERC8d: Wide dispersive outdoor use of processing aids in open systems

PROC1: Use in closed process, no likelihood of exposure

PROC10: Roller application or brushing

PROC11: Non-industrial spraying

PROC13: Treatment of articles by dipping and pouring

PROC15: Use as laboratory reagent

PROC19: Manual activities involving hand contact

PROC2: Use in closed, continuous process with occasional controlled exposure

PROC3: Use in closed batch process (synthesis or formulation)

PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

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PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Consumer application of coatings

Main User Groups : SU 21: Consumer uses: Private households (= general public = consumers)
Environmental Release Categories : ERC8d: Wide dispersive outdoor use of processing aids in open systems
Chemical product category : PC9a: Coatings and paints, thinners, paint removers
PC18: Ink and toners

2.1 Contributing scenario controlling environmental exposure for: ERC8d: Wide dispersive outdoor use of processing aids in open systems

Amount used

Regional use tonnage : 1 ton(s)/year
(tonnes/year):
Fraction of EU tonnage used in region: : 10 %
Fraction of Regional tonnage used locally: : 100 %
Maximum daily site tonnage : 0.273973 kg/day
(kg/day):

Environment factors not influenced by risk management

Dilution Factor (River) : 10
Dilution Factor (Coastal Areas) : 100

Other given operational conditions affecting environmental exposure

Number of emission days per year : 365
Emission or Release Factor: Air : 98.5 %
Emission or Release Factor: Water : 1 %
Emission or Release Factor: Soil : 0.5 %
Remarks : CEPE SPERC 8d.1a.v.1

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling consumer exposure for: PC9a: Coatings and paints, thinners, paint removers

Activity : Application

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 0.4%.

Amount used

Amount used : 1000 g
Remarks : Inhalation
Amount used : 0.5 g

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Remarks : Dermal

Frequency and duration of use

Frequency of use : 1 event/day
Remarks : Inhalation, Dermal
Exposure duration : 60 min
Remarks : Inhalation
Application duration : 60 min
Remarks : Inhalation

Human factors not influenced by risk management

Dermal exposure : 430 cm²

Other given operational conditions affecting consumers exposure

Outdoor / Indoor : Indoor
Room size : 20 m³
Temperature : 20 °C
Ventilation rate per hour : 2.5
Remarks : L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Do not touch eyes when using this product.

2.3 Contributing scenario controlling consumer exposure for: PC9a: Coatings and paints, thinners, paint removers

Activity : Application

Product characteristics

Concentration of the Substance in Mixture/Article : < 0.3% in formulation.

Amount used

Amount used : 550 g
Remarks : Inhalation

Frequency and duration of use

Frequency of use : 1 event/day
Remarks : Inhalation, Dermal
Exposure duration : 90 min
Remarks : Inhalation
Application duration : 90 min
Remarks : Inhalation
Release duration : 90 min
Remarks : Dermal

Human factors not influenced by risk management

Dermal exposure : 430 cm²

Other given operational conditions affecting consumers exposure

Outdoor / Indoor : Indoor
Room size : 58 m³
Temperature : 20 °C
Ventilation rate per hour : 0.5

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Remarks : L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Do not touch eyes when using this product.

2.4 Contributing scenario controlling consumer exposure for: PC18: Ink and toners

Activity : Use in Printing inks, Service life - consumers

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 3%.

Amount used

Amount used : 8.85 g
Remarks : Inhalation
Amount used : 200 g
Remarks : Dermal

Frequency and duration of use

Frequency of use : 1 event/day
Remarks : Inhalation, Dermal
Release duration : 720 min
Remarks : Inhalation

Human factors not influenced by risk management

Dermal exposure : 430 cm²

Other given operational conditions affecting consumers exposure

Outdoor / Indoor : Indoor
Room size : 58 m³
Temperature : 20 °C
Ventilation rate per hour : 0.5
Remarks : L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Do not touch eyes when using this product.

2.5 Contributing scenario controlling consumer exposure for: PC9a: Coatings and paints, thinners, paint removers

Activity : Application, Use in Paints/Coatings

Product characteristics

Concentration of the Substance in Mixture/Article : < 0.3% in formulation.

Amount used

Amount used : 1250 g
Remarks : Inhalation

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Frequency and duration of use

Frequency of use : 1 event/day
Remarks : Inhalation, Dermal
Exposure duration : 132 min
Remarks : Inhalation
Application duration : 120 min
Remarks : Inhalation
Release duration : 120 min
Remarks : Dermal

Human factors not influenced by risk management

Dermal exposure : 860 cm²

Other given operational conditions affecting consumers exposure

Outdoor / Indoor : Indoor
Room size : 20 m³
Temperature : 20 °C
Ventilation rate per hour : 0.6
Remarks : L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Do not touch eyes when using this product.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8d	EUSES		Fresh water		< 0.0001 mg/L	0.00027
			Fresh water sediment		0.00013 mg/kg dry weight	0.00027
			Marine water		< 0.0001 mg/L	0.00027
			Marine sediment		< 0.0001 mg/kg dry weight	0.00027
			Soil		< 0.0001 mg/kg dry weight	0.00069
			Sewage treatment plant		0.00017 mg/L	< 0.0001

Consumers

Contributing Scenario	Exposure Assessment	Specific conditions	Value	Level of Exposure	RCR
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	Method				
PC9a	"Consexpo"		Long term inhalation	1.391 mg/m3	0.69535
	"Consexpo"		Long term dermal	0.03333 mg/kg bw/day	0.13889
PC9a	"Consexpo"		Long term inhalation	0.04913 mg/m3	0.02457
	"Consexpo"		Long term dermal	0.135 mg/kg bw/day	0.5625
PC18	"Consexpo"		Long term inhalation	0.31804 mg/m3	0.15902
	"Consexpo"		Long term dermal	0.0065 mg/kg bw/day	0.02708
PC9a	"Consexpo"	Application, Use in Paints/Coatings	Long term inhalation	0.17391 mg/m3	0.08695
	"Consexpo"		Long term dermal	0.180 mg/kg bw/day	0.750

: Floor polish/sealer/stripper

: Use in Printing inks

: Paint strippers

: Service life - consumers

ERC8d: Wide dispersive outdoor use of processing aids in open systems

PC18: Ink and toners

PC9a: Coatings and paints, thinners, paint removers

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

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1. Short title of Exposure Scenario: Industrial uses, Use in oilfield applications

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)
Process categories	: PROC1: Use in closed process, no likelihood of exposure PROC2: Use in closed, continuous process with occasional controlled exposure PROC3: Use in closed batch process (synthesis or formulation) PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

2.1 Contributing scenario controlling environmental exposure for: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 2.857 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 350
Emission or Release Factor: Air	: 0.1 %
Emission or Release Factor: Water	: 0.05 %
Emission or Release Factor: Soil	: 0.1 %

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Air	: Waste gas treatment by thermal oxidation, Exhaust air scrubber
Water	: Aerobic biological treatment
Soil	: Sewage sludge incineration, No application of sewage sludge

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to soil, Sealing of all relevant soil surfaces in the facility is required

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC1: Use in closed process, no likelihood of exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

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Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

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2.5 Contributing scenario controlling worker exposure for: PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

Activity : General exposures, Batch process, With sample collection, Open systems, Industrial use

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear suitable coveralls to prevent exposure to the skin.
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.9 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment	Specific conditions	Compartment	Value	Level of Exposure	RCR
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	Method					
ERC4	EUSES		Fresh water		< 0.0001 mg/L	0.0002
			Fresh water sediment		< 0.0001 mg/kg dry weight	0.0002
			Marine water		< 0.0001 mg/L	0.0002
			Marine sediment		< 0.0001 mg/kg dry weight	0.0002
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC1	ECETOC TRA		Long term inhalation	0.00127 mg/m ³	0.00039
	ECETOC TRA		Long term dermal	< 0.0001 mg/kg bw/day	< 0.0001
PROC2	ECETOC TRA		Long term inhalation	0.12725 mg/m ³	0.03856
	ECETOC TRA		Long term dermal	0.00343 mg/kg bw/day	0.00312
PROC3	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.00171 mg/kg bw/day	0.00156
PROC4	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.01714 mg/kg bw/day	0.01558
PROC8a	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117
PROC8b	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.03429 mg/kg bw/day	0.03117

ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

PROC1: Use in closed process, no likelihood of exposure

PROC2: Use in closed, continuous process with occasional controlled exposure

PROC3: Use in closed batch process (synthesis or formulation)

PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

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PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Professional uses, Use in oilfield applications

Main User Groups	: SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
Environmental Release Categories	: ERC8d: Wide dispersive outdoor use of processing aids in open systems
Process categories	: PROC1: Use in closed process, no likelihood of exposure PROC2: Use in closed, continuous process with occasional controlled exposure PROC3: Use in closed batch process (synthesis or formulation) PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC8a: Transfer of substance or preparation (charging/discharging) from/ to vessels/ large containers at non-dedicated facilities

2.1 Contributing scenario controlling environmental exposure for: ERC8d: Wide dispersive outdoor use of processing aids in open systems

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 10 %
Fraction of Regional tonnage used locally:	: 0.2 %
Maximum daily site tonnage (kg/day):	: 0.000548 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 365
Emission or Release Factor: Air	: 1.5 %
Emission or Release Factor: Water	: 0.05 %
Emission or Release Factor: Soil	: 0.1 %
Remarks	: ESVOC SpERC 8.5b.v1

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
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Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant	: Municipal sewage treatment plant
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2.2 Contributing scenario controlling worker exposure for: PROC1: Use in closed process, no likelihood of exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

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Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC3: Use in closed batch process (synthesis or formulation)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

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Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.6 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 5%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

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Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8d	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001 mg/L	< 0.0001
			Marine sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC1	ECETOC TRA		Long term inhalation	0.00127 mg/m ³	0.00039
	ECETOC TRA		Long term dermal	< 0.0001 mg/kg bw/day	< 0.0001
PROC2	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.00343 mg/kg bw/day	0.00312
PROC3	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.00171 mg/kg bw/day	0.00156
PROC4	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.01714 mg/kg bw/day	0.01558
PROC8a	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term	0.03429	0.03117

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			dermal	mg/kg bw/day	
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ERC8d: Wide dispersive outdoor use of processing aids in open systems

PROC1: Use in closed process, no likelihood of exposure

PROC2: Use in closed, continuous process with occasional controlled exposure

PROC3: Use in closed batch process (synthesis or formulation)

PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users

http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Industrial uses, Adhesives, sealants

Main User Groups	: SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories	: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)
Process categories	: PROC1: Use in closed process, no likelihood of exposure PROC2: Use in closed, continuous process with occasional controlled exposure PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact) PROC7: Industrial spraying PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing) PROC10: Roller application or brushing PROC13: Treatment of articles by dipping and pouring PROC15: Use as laboratory reagent PROC17: Lubrication at high energy conditions in metal working operations PROC19: Manual activities involving hand contact

2.1 Contributing scenario controlling environmental exposure for: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 100 %
Fraction of Regional tonnage used locally:	: 100 %
Maximum daily site tonnage (kg/day):	: 4.545 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 220
Emission or Release Factor: Air	: 98 %
Emission or Release Factor: Water	: 2 %
Emission or Release Factor: Soil	: 0 %
Remarks	: ESVOC SpERC 4.3a.v1

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Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Air	: Waste gas treatment by thermal oxidation, Filtration, Adsorption, Exhaust air scrubber
Water	: Aerobic biological treatment
Soil	: No application of sewage sludge to soil, Sewage sludge incineration, Sealing of all relevant soil surfaces in the facility is required

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant	: Municipal sewage treatment plant
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2.2 Contributing scenario controlling worker exposure for: PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities

Activity	: Equipment cleaning and maintenance, Professional use
Product characteristics	
Concentration of the Substance in Mixture/Article	: Covers percentage substance in the product up to 30%.
Physical Form (at time of use)	: liquid
Frequency and duration of use	
Exposure duration	: < 480 min
Remarks	: Inhalation, Dermal
Frequency of use	: <= 240 days/year

Human factors not influenced by risk management

Dermal exposure	: Palm of one hand (240 cm ²)
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Other operational conditions affecting workers exposure

Outdoor / Indoor	: Indoor
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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

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Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

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2.6 Contributing scenario controlling worker exposure for: PROC7: Industrial spraying

Activity	: Option 1, Spraying (automatic/robotic)
Product characteristics	
Concentration of the Substance in Mixture/Article	: Covers percentage substance in the product up to 10%.
Physical Form (at time of use)	: liquid
Amount used	
Application rate	: 0.3 - 3 L/min
Frequency and duration of use	
Exposure duration	: < 480 min
Remarks	: Inhalation, Dermal
Frequency of use	: <= 240 days/year
Human factors not influenced by risk management	
Dermal exposure	: Both hands plus forearms (1500 cm2).
Other operational conditions affecting workers exposure	
Outdoor / Indoor	: Indoor
Room size	: 300 m3
Temperature	: 20 °C
Ventilation rate per hour	: 3
	: , Spraying with high compressed air use, Ensure that direction of application is only horizontal or downward.

Technical conditions and measures

Ensure fixed capturing hood is used. (Effectiveness (of a measure): 99 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear suitable coveralls to prevent exposure to the skin.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable face shield.

2.7 Contributing scenario controlling worker exposure for: PROC7: Industrial spraying

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Activity : Option 2, Manual spray application (liquid products)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Amount used

Application rate : 0.3 - 3 L/min

Frequency and duration of use

Exposure duration : < 240 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 300 m3
Temperature : 20 °C
Ventilation rate per hour : 3
: , Spraying with no or low compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
Gloves APF 20 (Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear suitable face shield.
Wear respiratory protection. (Effectiveness (of a measure): 90 %)

2.8 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

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Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 95 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.9 Contributing scenario controlling worker exposure for: PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

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Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.10 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 60 - 240 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.11 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.

Physical Form (at time of use) : liquid

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Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.12 Contributing scenario controlling worker exposure for: PROC15: Use as laboratory reagent

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

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Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

2.13 Contributing scenario controlling worker exposure for: PROC17: Lubrication at high energy conditions in metal working operations

Activity : Application of liquids in high speed processes (large scale)

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 10%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 240 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Room size : 300 m³

Temperature : > 20 °C

Ventilation rate per hour : 3

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

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Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear suitable coveralls to prevent exposure to the skin.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.14 Contributing scenario controlling worker exposure for: PROC19: Manual activities involving hand contact

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 10%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 15 - 60 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands and main part of arms (1980 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC4	EUSES		Fresh water		0.00058 mg/L	0.00685
			Fresh water sediment		0.00297 mg/kg dry	0.00685

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				weight	
			Marine water	< 0.0001 mg/L	0.00685
			Marine sediment	0.00030 mg/kg dry weight	0.00685
			Soil	0.00018 mg/kg dry weight	0.00502
			Sewage treatment plant	0.00575 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC1	ECETOC TRA		Long term inhalation	0.00764 mg/m ³	0.00231
	ECETOC TRA		Long term dermal	0.00051 mg/kg bw/day	0.00047
PROC2	ECETOC TRA		Long term inhalation	0.7635 mg/m ³	0.23136
	ECETOC TRA		Long term dermal	0.02057 mg/kg bw/day	0.01870
PROC4	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.10286 mg/kg bw/day	0.09351
PROC5	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.20571 mg/kg bw/day	0.18701
PROC7	ART	Spraying (automatic/robotic)	Long term inhalation	0.430 mg/m ³	0.13030
	ECETOC TRA		Long term dermal	0.21429 mg/kg bw/day	0.19481
PROC7	ART	Manual spray application (liquid products)	Long term inhalation	1.9 mg/m ³	0.57576
	ECETOC TRA		Long term dermal	0.10714 mg/kg bw/day	0.09740
PROC8b	ECETOC TRA		Long term inhalation	0.19087 mg/m ³	0.05784
	ECETOC TRA		Long term dermal	0.20571 mg/kg bw/day	0.18701
PROC9	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.10286 mg/kg bw/day	0.09351
PROC10	ECETOC TRA		Long term inhalation	1.527 mg/m ³	0.46273
	ECETOC TRA		Long term dermal	0.13714 mg/kg bw/day	0.12467

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PROC13	ECETOC TRA		Long term inhalation	0.7635 mg/m ³	0.23136
	ECETOC TRA		Long term dermal	0.20571 mg/kg bw/day	0.18701
PROC15	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.00514 mg/kg bw/day	0.00467
PROC17	ART		Long term inhalation	0.430 mg/m ³	0.13030
	ECETOC TRA		Long term dermal	0.13714 mg/kg bw/day	0.12467
PROC19	ECETOC TRA		Long term inhalation	0.509 mg/m ³	0.15424
	ECETOC TRA		Long term dermal	0.70714 mg/kg bw/day	0.64286

ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

PROC1: Use in closed process, no likelihood of exposure

PROC10: Roller application or brushing

PROC13: Treatment of articles by dipping and pouring

PROC15: Use as laboratory reagent

PROC17: Lubrication at high energy conditions in metal working operations

PROC19: Manual activities involving hand contact

PROC2: Use in closed, continuous process with occasional controlled exposure

PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises

PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)

PROC7: Industrial spraying

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
http://guidance.echa.europa.eu/downstream_users_en.htm

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1. Short title of Exposure Scenario: Professional uses, Adhesives, sealants

Main User Groups	: SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
Environmental Release Categories	: ERC8a: Wide dispersive indoor use of processing aids in open systems
Process categories	: PROC2: Use in closed, continuous process with occasional controlled exposure PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing) PROC10: Roller application or brushing PROC11: Non-industrial spraying PROC13: Treatment of articles by dipping and pouring PROC15: Use as laboratory reagent PROC17: Lubrication at high energy conditions in metal working operations PROC19: Manual activities involving hand contact

2.1 Contributing scenario controlling environmental exposure for: ERC8a: Wide dispersive indoor use of processing aids in open systems

Amount used

Regional use tonnage (tonnes/year):	: 1 ton(s)/year
Fraction of EU tonnage used in region:	: 10 %
Fraction of Regional tonnage used locally:	: 0.050 %
Maximum daily site tonnage (kg/day):	: 0.000137 kg/day

Environment factors not influenced by risk management

Dilution Factor (River)	: 10
Dilution Factor (Coastal Areas)	: 100

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 365
Emission or Release Factor: Air	: 98 %
Emission or Release Factor: Water	: 1 %
Emission or Release Factor: Soil	: 1 %
Remarks	: ESVOC SpERC 8.3b.v1

Technical conditions and measures / Organizational measures

Compartment	: Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Water	: Aerobic biological treatment

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Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC2: Use in closed, continuous process with occasional controlled exposure

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

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Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

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Conditions and measures related to personal protection, hygiene and health evaluation

Provide adequate information, instruction and training for operators.

In case of exposure:

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Use suitable eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC10: Roller application or brushing

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear respiratory protection. (Effectiveness (of a measure): 90 %)

Wear suitable face shield.

2.6 Contributing scenario controlling worker exposure for: PROC11: Non-industrial spraying

Activity : Manual spray application (liquid products)

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.

Physical Form (at time of use) : liquid

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Amount used

Application rate : 0.3 - 3 L/min

Frequency and duration of use

Exposure time : < 120 min
Frequency of use : minutes/day
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands plus forearms (1500 cm2).

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 100 m3
Temperature : 20 °C
Ventilation rate per hour : 1
: , Spraying with no or low compressed air use, Ensure that direction of application is only horizontal or downward.

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear respiratory protection. (Effectiveness (of a measure): 95 %)
Wear suitable face shield.

2.7 Contributing scenario controlling worker exposure for: PROC13: Treatment of articles by dipping and pouring

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 30%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palms of both hands (480 cm2)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

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Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear suitable face shield.

2.8 Contributing scenario controlling worker exposure for: PROC15: Use as laboratory reagent

Product characteristics

Concentration of the Substance in : Covers percentage substance in the product up to 30%.

Mixture/Article

Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 480 min

Remarks : Inhalation, Dermal

Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

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Use suitable eye protection.

2.9 Contributing scenario controlling worker exposure for: PROC17: Lubrication at high energy conditions in metal working operations

Activity : Option 1, Application of liquids in high speed processes, Small scale, Manual process

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 240 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 100 m³
Temperature : > 20 °C
Ventilation rate per hour : 1

Technical conditions and measures

Ensure fixed capturing hood is used. (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Wear respiratory protection. (Effectiveness (of a measure): 90 %)
Use suitable eye protection.

2.10 Contributing scenario controlling worker exposure for: PROC17: Lubrication at high energy conditions in metal working operations

Activity : Option 2, Application of liquids in high speed processes, Small scale, Manual process

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Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : < 90 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands (960 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor
Room size : 100 m³
Temperature : > 20 °C
Ventilation rate per hour : 1

Technical conditions and measures

Ensure fixed capturing hood is used. (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented
Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training
Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.
(Effectiveness (of a measure): 95 %)
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.11 Contributing scenario controlling worker exposure for: PROC19: Manual activities involving hand contact

Product characteristics

Concentration of the Substance in Mixture/Article : Covers percentage substance in the product up to 10%.
Physical Form (at time of use) : liquid

Frequency and duration of use

Exposure duration : 60 - 240 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Both hands and main part of arms (1980 cm²)

Other operational conditions affecting workers exposure

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Outdoor / Indoor

: Indoor

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Wear respiratory protection. (Effectiveness (of a measure): 90 %)

Wear suitable face shield.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8a	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001 mg/L	< 0.0001
			Marine sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC2	ECETOC TRA		Long term inhalation	0.7635 mg/m ³	0.23136
	ECETOC TRA		Long term dermal	0.02057 mg/kg bw/day	0.01870
PROC8b	ECETOC TRA		Long term inhalation	0.7635 mg/m ³	0.23136
	ECETOC TRA		Long term	0.20571	0.18701

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			dermal	mg/kg bw/day	
PROC9	ECETOC TRA		Long term inhalation	1.527 mg/m ³	0.46273
	ECETOC TRA		Long term dermal	0.10286 mg/kg bw/day	0.09351
PROC10	ECETOC TRA		Long term inhalation	0.63625 mg/m ³	0.19280
	ECETOC TRA		Long term dermal	0.13714 mg/kg bw/day	0.12467
PROC11	ART	Manual spray application (liquid products)	Long term inhalation	1.75 mg/m ³	0.53030
	ECETOC TRA		Long term dermal	0.13393 mg/kg bw/day	0.12175
PROC13	ECETOC TRA		Long term inhalation	1.527 mg/m ³	0.46273
	ECETOC TRA		Long term dermal	0.20571 mg/kg bw/day	0.18701
PROC15	ECETOC TRA		Long term inhalation	0.7635 mg/m ³	0.23136
	ECETOC TRA		Long term dermal	0.00514 mg/kg bw/day	0.00467
PROC17	ART	Option 1	Long term inhalation	1.4 mg/m ³	0.42424
	ECETOC TRA		Long term dermal	0.13714 mg/kg bw/day	0.12467
PROC17	ART	Option 2	Long term inhalation	2.7 mg/m ³	0.81818
	ECETOC TRA		Long term dermal	0.02571 mg/kg bw/day	0.02338
PROC19	ECETOC TRA		Long term inhalation	0.38175 mg/m ³	0.11568
	ECETOC TRA		Long term dermal	0.70714 mg/kg bw/day	0.64286

ERC8a: Wide dispersive indoor use of processing aids in open systems

PROC10: Roller application or brushing

PROC11: Non-industrial spraying

PROC13: Treatment of articles by dipping and pouring

PROC15: Use as laboratory reagent

PROC17: Lubrication at high energy conditions in metal working operations

PROC19: Manual activities involving hand contact

PROC2: Use in closed, continuous process with occasional controlled exposure

PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities

PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For further information, please also consult our Internet site: Downstream Users
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1. Short title of Exposure Scenario: Use in adhesives and sealants (Consumer)

Main User Groups : SU 21: Consumer uses: Private households (= general public = consumers)
Environmental Release Categories : ERC8a: Wide dispersive indoor use of processing aids in open systems
Chemical product category : PC1: Adhesives, sealants

2.1 Contributing scenario controlling environmental exposure for: ERC8a: Wide dispersive indoor use of processing aids in open systems

Amount used

Regional use tonnage : 1 ton(s)/year
(tonnes/year):
Fraction of EU tonnage used in : 10 %
region:
Fraction of Regional tonnage used : 0.050 %
locally:
Maximum daily site tonnage : 0.000137 kg/day
(kg/day):

Environment factors not influenced by risk management

Dilution Factor (River) : 10
Dilution Factor (Coastal Areas) : 100

Other given operational conditions affecting environmental exposure

Number of emission days per year : 365
Emission or Release Factor: Air : 98 %
Emission or Release Factor: Water : 1 %
Emission or Release Factor: Soil : 1 %
Remarks : ESVOC SpERC 8.3b.v1

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling consumer exposure for: PC1: Adhesives, sealants

Activity : Application

Amount used

Amount used : 22000 g
Remarks : Inhalation

Frequency and duration of use

Frequency of use : 1 event/day
Remarks : Inhalation, Dermal
Exposure duration : 480 min

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Remarks	: Inhalation
Application duration	: 480 min
Remarks	: Inhalation
Release duration	: 480 min
Remarks	: Dermal

Human factors not influenced by risk management

Dermal exposure	: 430 cm2
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Other given operational conditions affecting consumers exposure

Outdoor / Indoor	: Indoor
Room size	: 58 m3
Temperature	: 20 °C
Ventilation rate per hour	: 2.5
Remarks	: L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures	: Do not touch eyes when using this product.
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2.3 Contributing scenario controlling consumer exposure for: PC1: Adhesives, sealants

Activity	: Application
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Amount used

Amount used	: 750 g
Remarks	: Inhalation
Amount used	: 0.5 g
Remarks	: Dermal

Frequency and duration of use

Frequency of use	: 1 event/day
Remarks	: Inhalation, Dermal
Exposure duration	: 240 min
Remarks	: Inhalation
Application duration	: 240 min
Remarks	: Inhalation

Human factors not influenced by risk management

Dermal exposure	: 110 cm2
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Other given operational conditions affecting consumers exposure

Outdoor / Indoor	: Indoor
Room size	: 58 m3
Temperature	: 20 °C
Ventilation rate per hour	: 2.5
Remarks	: L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures	: Do not touch eyes when using this product.
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2.4 Contributing scenario controlling consumer exposure for: PC1: Adhesives, sealants

Activity : Application

Amount used

Amount used : 9000 g
Remarks : Inhalation

Frequency and duration of use

Frequency of use : 1 event/day
Remarks : Inhalation, Dermal
Exposure duration : 75 min
Remarks : Inhalation
Application duration : 75 min
Remarks : Inhalation
Release duration : 75 min
Remarks : Dermal

Human factors not influenced by risk management

Dermal exposure : 110 cm²

Other given operational conditions affecting consumers exposure

Outdoor / Indoor : Indoor
Room size : 58 m³
Temperature : 20 °C
Ventilation rate per hour : 2.5
Remarks : L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Do not touch eyes when using this product.

2.5 Contributing scenario controlling consumer exposure for: PC1: Adhesives, sealants

Activity : Mixing and loading

Amount used

Amount used : 20 g
Remarks : Inhalation
Amount used : 0.050 g
Remarks : Dermal

Frequency and duration of use

Frequency of use : 1 event/day
Remarks : Inhalation, Dermal
Exposure duration : 5 min
Remarks : Inhalation
Application duration : 5 min
Remarks : Inhalation

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Human factors not influenced by risk management

Dermal exposure : 2 cm²

Other given operational conditions affecting consumers exposure

Outdoor / Indoor : Indoor
Room size : 20 m³
Temperature : 20 °C
Ventilation rate per hour : 2.5
Remarks : L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Do not touch eyes when using this product.

2.6 Contributing scenario controlling consumer exposure for: PC1: Adhesives, sealants

Activity : Application

Amount used

Amount used : 20 g
Remarks : Inhalation
Amount used : 0.100 g
Remarks : Dermal

Frequency and duration of use

Frequency of use : 1 event/day
Remarks : Inhalation, Dermal
Exposure duration : 120 min
Remarks : Inhalation
Application duration : 30 min
Remarks : Inhalation

Human factors not influenced by risk management

Dermal exposure : 43 cm²

Other given operational conditions affecting consumers exposure

Outdoor / Indoor : Indoor
Room size : 20 m³
Temperature : 20 °C
Ventilation rate per hour : 2.5
Remarks : L/h

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Do not touch eyes when using this product.

3. Exposure estimation and reference to its source

Environment

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Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8a	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001 mg/L	< 0.0001
			Marine sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

Consumers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PC1	"Consexpo"		Long term inhalation	0.57757 mg/m3	0.28878
	"Consexpo"		Long term dermal	0.120 mg/kg bw/day	0.500
PC1	"Consexpo"		Long term inhalation	1.163 mg/m3	0.58172
	"Consexpo"		Long term dermal	0.050 mg/kg bw/day	0.20833
PC1	"Consexpo"		Long term inhalation	0.93262 mg/m3	0.46631
	"Consexpo"		Long term dermal	0.09375 mg/kg bw/day	0.39063
PC1	"Consexpo"		Long term inhalation	0.00024 mg/m3	0.00012
	"Consexpo"		Long term dermal	0.04167 mg/kg bw/day	0.17361
PC1	"Consexpo"		Long term inhalation	0.77522 mg/m3	0.38761
	"Consexpo"		Long term dermal	0.08333 mg/kg bw/day	0.34722

: Two-component glue

: Wood parquet glue

: Carpet glue

ERC8a: Wide dispersive indoor use of processing aids in open systems

PC1: Adhesives, sealants

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4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

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1. Short title of Exposure Scenario: Laboratory Reagents

Main User Groups : SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
Environmental Release Categories : ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)
Process categories : PROC15: Use as laboratory reagent

2.1 Contributing scenario controlling environmental exposure for: ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

Amount used

Regional use tonnage : 1 ton(s)/year
(tonnes/year):
Fraction of EU tonnage used in : 100 %
region:
Fraction of Regional tonnage used : 100 %
locally:
Maximum daily site tonnage : 50 kg/day
(kg/day):

Environment factors not influenced by risk management

Dilution Factor (River) : 10
Dilution Factor (Coastal Areas) : 100

Other given operational conditions affecting environmental exposure

Number of emission days per year : 20
Emission or Release Factor: Air : 2.5 %
Emission or Release Factor: Water : 2 %
Emission or Release Factor: Soil : 0.01 %
Remarks : ESVOC SPERC 4.24.v1

Technical conditions and measures / Organizational measures

Compartment : Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant
Water : Aerobic biological treatment

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC15: Use as laboratory reagent

Product characteristics

Concentration of the Substance in : Covers the percentage of the substance in the product up to
Mixture/Article 100 % (unless stated differently).
Physical Form (at time of use) : liquid

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Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 90 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

Gloves APF 20 (Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC4	EUSES		Fresh water		0.00633 mg/L	0.07446
			Fresh water sediment		0.03228 mg/kg dry weight	0.07437
			Marine water		0.00063 mg/L	0.07446
			Marine sediment		0.00323 mg/kg dry weight	0.07437
			Soil		< 0.0001 mg/kg dry weight	0.00065
			Sewage treatment plant		0.06321 mg/L	0.00063

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Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC15	ECETOC TRA		Long term inhalation	1.273 mg/m ³	0.38561
	ECETOC TRA		Long term dermal	0.01714 mg/kg bw/day	0.01558

ERC4: Use of non-reactive processing aid at industrial site (no inclusion into or onto article)

PROC15: Use as laboratory reagent

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

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1. Short title of Exposure Scenario: Laboratory Reagents

Main User Groups : SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)

Environmental Release Categories : ERC8a: Wide dispersive indoor use of processing aids in open systems

Process categories : PROC15: Use as laboratory reagent

2.1 Contributing scenario controlling environmental exposure for: ERC8a: Wide dispersive indoor use of processing aids in open systems

Amount used

Regional use tonnage : 1 ton(s)/year
(tonnes/year):

Fraction of EU tonnage used in : 10 %
region:

Fraction of Regional tonnage used : 0.050 %
locally:

Maximum daily site tonnage : 0.000137 kg/day
(kg/day):

Environment factors not influenced by risk management

Dilution Factor (River) : 10

Dilution Factor (Coastal Areas) : 100

Other given operational conditions affecting environmental exposure

Number of emission days per year : 365

Emission or Release Factor: Air : 50 %

Emission or Release Factor: Water : 50 %

Emission or Release Factor: Soil : 0 %

Remarks : ESVOC SPERC 8.17.v1

Technical conditions and measures / Organizational measures

Compartment : Fresh water, Fresh water sediment, Marine water, Marine sediment, Soil, Sewage treatment plant

Water : Aerobic biological treatment

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant : Municipal sewage treatment plant

2.2 Contributing scenario controlling worker exposure for: PROC15: Use as laboratory reagent

Product characteristics

Concentration of the Substance in : Covers the percentage of the substance in the product up to
Mixture/Article 100 % (unless stated differently).

Physical Form (at time of use) : liquid

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Frequency and duration of use

Exposure duration : < 480 min
Remarks : Inhalation, Dermal
Frequency of use : <= 240 days/year

Human factors not influenced by risk management

Dermal exposure : Palm of one hand (240 cm²)

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor

Technical conditions and measures

Local exhaust ventilation (Effectiveness (of a measure): 80 %)

Organisational measures to prevent /limit releases, dispersion and exposure

Assumes a good basic standard of occupational hygiene is implemented

Supervision in place to check that the RMMs in place are being used correctly and OCs followed.

Conditions and measures related to personal protection, hygiene and health evaluation

Specific worker training

Wear chemically resistant gloves (tested to EN374) in combination with specific activity training.

(Effectiveness (of a measure): 95 %)

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR
ERC8a	EUSES		Fresh water		< 0.0001 mg/L	< 0.0001
			Fresh water sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Marine water		< 0.0001 mg/L	< 0.0001
			Marine sediment		< 0.0001 mg/kg dry weight	< 0.0001
			Soil		< 0.0001 mg/kg dry weight	0.00054
			Sewage treatment plant		< 0.0001 mg/L	< 0.0001

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Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR
PROC15	ECETOC TRA	Professional use	Long term inhalation	0.76 mg/m ³	0.23
			Long term dermal	< 0.01 mg/kg bw/day	< 0.01

ERC8a: Wide dispersive indoor use of processing aids in open systems

PROC15: Use as laboratory reagent

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